Ruogu "Rena" Fang MS, Ph.D.

Professor, Flowers Family Dean's Faculty Fellow

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EXECUTIVE SUMMARY:

Ruogu "Rena" Fang, M.S., Ph.D., is a tenured Professor and Flowers Family Dean's Faculty Fellow in the Biomedical Engineering Department at Vanderbilt University. She is the Director of the Smart Medical Informatics Learning and Evaluation (SMILE) Lab. She has been a Visiting Associate Professor at Stanford University sponsored by the Stanford Science Visiting Professorship, Harvard University, Mass General Brighton (MGB), the University of Tokyo, and RIKEN.

Dr. Fang has established an internationally recognized research and educational program focused on Medical Artificial Intelligence (AI) for diagnosis and intervention of brain disorders, with a particular emphasis on neurodegenerative diseases. Her research themes are artificial intelligence (AI)-empowered precision brain health and brain/bio-inspired AI. Her transdisciplinary research impact has spanned from creating a new modularized machine learning system for early diagnosis of neurodegenerative diseases using novel imaging to individualized intervention using non-invasive brain stimulation to prevent dementia. Her lab has been continuously funded by NSF, NIH, AFRL, and industry since 2016. Dr. Fang serves as the Principal Investigator (PI) and Co-Investigator on 41 funded grants, among which she served as PI on 19 grants, with a total funding of $55M, and PI of $10M. She has ten issued patents. She was selected as the National Academy of Medicine's (NAM's) Victor J. Dzau Emerging Leaders in Health and Medicine (ELHM) Scholar, Rising Stars by the Academy of Science, Engineering, and Medicine of Florida (ASEMFL), and received the Best Paper Award from the IEEE International Conference on Image Processing, UF Inaugural Artificial Intelligence (AI) Course Award, University of Florida College of Engineering Faculty Award for Excellence in Innovation, Faculty Doctoral Dissertation Advisor/Mentoring Award, UF BME Faculty Research Excellence Award, among others. She has 180+ peer-reviewed publications, including highly respected journals and top conferences of *The Lancet Digital Health, JAMA Neurology, Nature Computational Sciences, Nature Communications, PNAS Nexus, Nature npj Digital Medicine, npj Women's Health, Medical Image Analysis, IEEE Transactions on Medical Imaging, ICCV, MICCAI*, that are well cited (h-index=31, I_{10}-index=63, 5500+ citations) and frequently featured by international major media (e.g., *Forbes Magazine, The Washington Post, ABC, National Geography, The Hindu, EveSmart, and Eminetra*). She is ranked among the **Top 1%** of researchers worldwide in image analysis.

Dr. Fang is passionate about educating the next generation of transdisciplinary biomedical researchers from diverse backgrounds, particularly those from underrepresented and marginalized communities. She is the inaugural chair of the Public Engagement and Outreach Committee and was instrumental in translating the Shepard Broad Foundation Community Outreach Fund into society-benefiting actions. In her laboratory, she has mentored 3 Postdoc Associates, 13 doctoral and 20+ master's graduate students, and over 60 undergraduates. Many of her students have gone on to secure prestigious fellowships, such as NIH F31, T32, TL1, R25, and NSF GRFP fellowships, Best Paper Awards, and became tenure-track faculty position at top institutions (e.g., Mayo Clinic) or high-tech companies (e.g., Meta, Microsoft, Amazon).

Dr. Fang contributes to building the next generation of leaders in medical AI by teaching computer applications in BME. She has introduced innovative, high-value educational pedagogies in her undergraduate and graduate classrooms during the pandemic, which have led to *the best offering* of these classes in terms of student evaluation in the history of the BME department. She also created a new AI course series, "Medical Artificial Intelligence (MAI)," and received the inaugural AI Course Award from UF and the BME Departmental Faculty Excellence in Teaching Award.

Dr. Fang has enthusiastically promoted the emerging field of Medical AI while faithfully serving the profession. She is the President of Women in MICCAI and a mentor of AnitaBorg.org and the National Academy of Inventors Global Academy Inventors Networks (GAIN). Dr. Fang currently serves as the Associate Editor of *Medical Image Analysis*, serves as the inaugural Topic Editor of "*Women in Brain Imaging and Brain Stimulation*" in *Frontiers of Human Neuroscience*, and organized a special issue in *Computerized Medical Imaging and Graphics*. She chaired 10+ workshops, sessions, and tracks at international conferences. Dr. Fang has regularly served on NIH study sections and NSF panels (10+ years, 90+ grant proposals) and reviewed for 30+ journals (100+ manuscripts/books).

EDUCATION

2014 **Ph.D. in Electrical and Computer Engineering**, Computer Science Minor, Cornell University, Ithaca, NY

2009 **B.E. Information Engineering,** Rank 1st in Chu-Kochen Honors College, Zhejiang University, China

2009 **Visiting Researcher**, Chemical Engineering and Biotechnology, The University of Cambridge, Cambridge, UK

2008 **Exchange Student**, Electrical and Electronic Engineering, University of Hong Kong, Hong Kong

PROFESSIONAL EXPERIENCE

8/16/2026 – Present	Professor, Biomedical Engineering, Vanderbilt University
	Flowers Family Dean's Faculty Fellow, Vanderbilt University
	Professor, Electrical and Computer Engineering, Vanderbilt University (affiliated)
	Professor, Department of Biomedical Informatics, Vanderbilt Health (affiliated)
	Professor, Department of Radiology, Vanderbilt Health (affiliated)
	Professor, Vanderbilt University Institute of Imaging and Science (VUIIS) (member)
	Professor, Vanderbilt Institute of Nanoscale Science and Engineering (VINSE) (member)
	Professor, Center for Computational System Biology (CCSB) (member)
6/1/2025 – 7/31/2025	Visiting Associate Professor, School of Engineering and Applied Sciences, Harvard University
6/1/2025 – 7/31/2025	Visiting Associate Professor, Harvard Medical School and Mass General Brigham
11/1/2024 – 02/28/2025	Stanford Science Visiting Professor, Stanford University
04/2025 – 05/2025	Visiting Associate Professor, the University of Tokyo & RIKEN
8/16/2026	Professor, J. Crayton Pruitt Family Biomedical Engineering Department, University of Florida
8/16/2022 – 8/15/2026	Associate Professor (with tenure), J. Crayton Pruitt Family Biomedical Engineering Department, University of Florida
	Affiliated Faculty, Norman Fixel Institute for Neurological Diseases
	Affiliated Faculty, Electrical and Computer Engineering Department
	Affiliated Faculty, Computer Information & Science & Engineering Department
	Affiliated Faculty Center for Cognitive Aging and Memory Clinical Translational Research (CAM)
	Affiliated Faculty, University of Florida Genetics Institute (UFGI)
	Affiliated Faculty, Emerging Pathogens Institute
	Affiliated Faculty, UF Health Cancer Center
	Affiliated Faculty, Radiology
01/01/2023 – 12/31/2025	Pruitt Family Endowed Faculty Fellow in Biomedical Engineering
10/2023 – 09/2025	President, Women in MICCAI (WiM), the Medical Image Computing and Computer Assisted Intervention (MICCAI)
03/01/2022 – Present	Associate Director, Intelligent Critical Care Center (IC^3), University of Florida
08/16/2017 – 08/15/2022	Assistant Professor, Biomedical Engineering Department, University of Florida
8/16/2014 – 8/15/2017	Assistant Professor, School of Computing and Information Sciences, Florida International University

HONORS AND DISTINCTIONS

External Faculty Awards

- National Academy of Medicine's (NAM's) Victor J. Dzau Emerging Leaders in Health and Medicine (ELHM) Scholar, 2026-2029 (only 10 nationwide)
- National Academy of Engineering (NAE) The Grainger Foundation Frontiers of Engineering 2026 Symposium, 2026
- Stanford Science Visiting Professorship, 2024-2025

- Senior Member, National Academy of Inventors (NAI), 2024
- Rising Stars (Engineering). the Academy of Science, Engineering, and Medicine of Florida (ASEMFL) Annual Meeting, 2023.
- Inaugural Class of the ACM Future of Computing Academy, Academy of Computing Machinery, 2017
- Ralph Lowe Junior Faculty Enhancement Award, Oak Ridge Associated Universities (1 of 35 awardees nationwide in the United States), 2016
- Robin Sidhu Memorial Young Scientist Award, Society of Brain Mapping and Therapeutics, 2016
- National Science Foundation CISE Research Initiation Initiative (CRII) Award, 2015
- Senior Member, Institute of Electrical and Electronics Engineers (IEEE), 2018

Internal Faculty Award

- Flowers Family Dean's Faculty Fellow, Vanderbilt University, 2026
- BME Shepard Broad Foundation Community Outreach Fund, University of Florida, 2026
- Promoted to Full Professor, University of Florida & Vanderbilt University, 2026
- Research Promotion Initiative (RPI) Award $1000, University of Florida, 2026
- Doctoral Dissertation Advisor/Mentoring Award, Herbert Wertheim College of Engineering, 2025
- Faculty Teaching Excellence Award, J. Crayton Pruitt Family Department of Biomedical Engineering, 2024
- Pioneering Research HiPerGator Award, University of Florida, 2024
- Inaugural AI Course Award, University of Florida, 2024
- Tenure and Promotion to Associate Professor, University of Florida, 2022.
- The University of Florida Herbert Wertheim College of Engineering 2022 Faculty Award for Excellence in Innovation, 2022.
- The University of Florida, J. Crayton Pruitt Family Department of Biomedical Engineering Faculty Research Excellence Award, 2021
- Great Teaching Certificate, University of Florida Center for Teaching Excellence, 2020
- Master Mentor Certificate, UF Training and Professional Development Program, Mentor Academy 2022

Best Paper Awards, Competition Awards, Reviewer Awards

- Best Paper Award, The 17th International Conference on Health Informatics (HEALTHINF/BIOSTEC), 2024
- Best Paper Presentation Award Runner-Up, Women in MICCAI, Sep. 2022 in conjunction with the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2022
- 1st Place at 2022 Georgia Tech GPU Hackathon
- 1st Place at the 2022 University of Florida GPU Hackathon
- UNIQUE-IVADO Award for Best Abstract (Undergraduate and Graduate Level) at Montreal AI& Neuroscience (MAIN) Conference, 2021
- David Goldstein M.D., Ph.D., Presidential Trainee Award, American Society for Clinical Pharmacology and Therapeutics Annual Meeting in Houston, TX, March 18-21, 2020. (For the highest-scoring abstract, Co-Authored Paper)
- IEEE Transaction on Medical Imaging (TMI) Distinguished Reviewer Bronze Level, 2021
- Top-ranked (#1) in "Fovea Detection" System, First International Diabetic Retinopathy Grading and

Segmentation Challenge, IEEE International Symposium of Biomedical Imaging, 2018

- Top-ranked (#1) in "Optical Disc Segmentation" System, First International Diabetic Retinopathy Grading and Segmentation Challenge, IEEE International Symposium of Biomedical Imaging, 2018
- Most Cited Paper since 2018, Medical Image Analysis, 2021
- Hottest Article in Medical Image Analysis, Elsevier Publisher, April - September 2014
- Best Paper Award, the 17th International Conference on Image Processing, First Author, 2010

Graduate School Awards

- Hsien Wu and Daisy Yen Wu Memorial Award (awarded to 5 outstanding graduate students at Cornell University), 2014
- Irwin and Joan Jacobs Fellowship, Cornell University, 2009
- Best Ph.D. Poster Award, Cornell Engineering Research Conference, 2010
- Student Travel Award, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2014
- Student Travel Award, the 17th International Conference on Image Processing, 2010
- IBM Cornell ECE Women Conference Travel Grant, the 15th International Conference on Medical Image Computing and Computer Assisted Intervention, 2012

Undergraduate School Awards

- Bao-Steel Scholarship, Zhejiang University, 2008 (Awarded at graduation to top one EE student)
- Li & Fung Scholarship, University of Hong Kong (full stipend and tuition for exchange students), 2007
- First Class Scholarship, Zhejiang University, 2006-2009
- Dean's List (top 1%), Zhejiang University, 2006-2009
- First Prize in the National Olympiad in Mathematics, 2001

AWARDS RECEIVED BY STUDENT ADVISEES

1. Olivia Ortlieb, IBM Fellowship, Vanderbilt University, 2026
2. Wasif Khan, Excellence in Research Award for Postdocs, University of Florida, 2026
3. Seowung Leem, Gator Engineering Attribute Award for Creativity, Herbert Wertheim College of Engineering, University of Florida, 2026
4. Gavin Hart, Second Place in Center for Cognitive Aging and Memory (CAM) Research Day, 2025
5. Skylar Stolte, Association for Academic Women's (AAW) Emerging Scholar Award, Madelyn Lockhart Dissertation Awards Committee, University of Florida, 2025
6. Skylar Stolte, Graduate Student Excellence Award, J. Crayton Pruitt Family Department of Biomedical Engineering, 2024
7. Joseph Cox, Supervised Teaching Excellence Award, J. Crayton Pruitt Family Department of Biomedical Engineering, 2024
8. Chaoyue Sun, International Student Achievement Awards Certificate of Outstanding Merit, International Center, University of Florida, 2024
9. Seowung Leem, International Student Achievement Awards Certificate of Outstanding Merit, International Center, University of Florida, 2024
10. Seowung Leem, Pruitt Research Day Best Poster Presentation Award, 2024

11. Skylar Stolte, the Center for Cognitive Aging and Memory (CAM) Dissertation Award, $2500, 2024
12. Gavin Hart, NIH R25-funded postbaccalaureate training program **G**ators **A**dvancing **T**hrough **O**pportunities in **R**esearch for **A**ging and **A**lzheimer's **D**isease **E**ducation (GATORAADE) Fellow, 2024
13. Skylar Stolte, the Center for Cognitive Aging and Memory (CAM) Student Travel Award, $1000, 2024
14. **Skylar Stolte, NIH F31 Ruth L. Kirschstein Predoctoral Individual National Research Service Award**, 2024
15. Seowumg Leem: IEEE EMBC NextGen Scholar Award, IEEE Engineering in Medicine and Biology Society (EMBS), 2024
16. **Joseph Cox: NIH NIAAA Kirschstein-Institutional National Research Service Award T32 Fellowship**, 2024
17. Skylar Stolte: Winner of Gator Engineering Attribute Award for Creativity, Herbert Wertheim College of Engineering, University of Florida, 2024
18. Skylar Stolte: Winner of the Leighton E. Cluff Award for Aging Research, University of Florida, 2024 ($1250).
19. **Chaoyue Sun, Best Paper Award (1 among all submissions),** The 17th International Conference on Health Informatics (HEALTHINF/BIOSTEC), Rome - Italy, published in Proceedings of the 17th International Joint Conference on Biomedical Engineering Systems and Technologies, 2024
20. Skylar E. Stolte: Second Place, WiM Inspirational Leadership Legacy (WILL), International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2023 ($300).
21. **Daniel Rodriguez: NIH T1D Kirschstein-Institutional National Research Service Award T32 Fellowship**, 2023
22. Grace Cheng & Akshay Ashok: UF AI Scholar, 2023
23. Skylar E. Stolte: Best Paper Presentation Award Runner Up, Women in MICCAI (WiM), International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022.
24. **Garrett Fullerton: NSF Graduate Research Fellowship Program (GRFP),** 2022
25. Garrett Fullerton: Outstanding Research Award, BME Undergraduate Research Day, 2022
26. Peng Liu: Best Abstract Award (Undergraduate and Graduate Level), Montreal AI & Neuroscience (MAIN) Conference, 2021
27. Charlie Tran: NextProf Pathfinder, University of Michigan and University of California, San Diego, 2021
28. Gianna Sweeting: John & Mittie Collins Engineering Scholarship, 2021
29. Charlie Tran: Ronald E. McNair Graduate Assistantship, 2021
30. **Yao Xiao: BMES Career Development Award, Biomedical Engineering Society (BMES),** 2020
31. **Yao Xiao: Graduate Student Speaker at the College of Engineering Commencement**, 2020
32. Peng Liu: UFII Graduate Student Fellowship, University of Florida Informatics Institute, 2020
33. Gianna Sweeting: Herbert Wertheim College of Engineering Scholarship, University of Florida, 2020
34. Charlie Tran: Ronald E. McNair Post-Baccalaureate Achievement Program, 2020
35. Gianna Sweeting: Fernandez Family Scholar, College of Engineering, University of Florida, 2020
36. Gianna Sweeting: University Scholar Program, University of Florida, 2020
37. Skylar E. Stolte: Graduate School Preeminence Award (GSPA), University of Florida, 2019
38. Yao Xiao: National Institutes of Health, National Institute of Biomedtylical Imaging and Bioengineering (NIBIB), National Cancer Institute (NCI), and Graduate Student Council Travel Awards, IEEE International Symposium on Biomedical Imaging (ISBI), 2020
39. Garrett Fullerton: University Scholar Program, 2020
40. **Maximillian Diaz: NSF Graduate Research Fellowship Program (GRFP),** 2020

41. **Daniel El Basha: NSF Graduate Research Fellowship Program (GRFP**), 2020
42. Kyle B. See: Graduate School Preeminence Award (GSPA), University of Florida, 2019
43. Kyle B. See: NIH CTSI TL1 Pre-doctoral Fellowship, 2019
44. Maximillian Diaz, University Scholar Program, 2019
45. Yao Xiao: Richard Tapia Celebration of Diversity in Computing Conference Scholarship, Academy of Computing Machinery (ACM), 2018
46. Yao Xiao, ACM/CHASE Student Travel Award, Connected Health: Applications, Systems and Engineering Technologies (CHASE), 2017
47. Yao Xiao, Student Travel Award (award rate=6%), International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2017
48. Akash Mathavan, University Scholar Program, 2018
49. Akshay Mathavan, University Scholar Program, 2018
50. Micheal Adeyosoye, McKnight Bridge Fellowship, 2015

GRANTS & FUNDING

Awarded Summary (2014-2024)	
Number of Grants:	41
Number of PI Grants:	19
Total Amount:	$ 56 M
PI Amount:	$ 10.5 M
Faculty Share:	$ 6 M

FEDERAL GRANTS

G1. NIH NIMH, R01MH112558, **MPI**, **$3,456,966**, 2026-2031. "MyAPS: An AI Platform for Generating Emotional Pictures." (**2% score)**

G2. NSF BCS- 2617585, **Co-PI** (PI: Odegaard) **$890,113 (Fang: $349,095),** 2026-2029. "Understanding the Role of Metacognitive Sensitivity in Human-AI Joint Decision-Making."

G3. NIH NIA, RF1/R01AG071469, **contact PI/MPI (Fang, Woods)**, **$2,925,577 (Fang: $1,393,011)**, 2021-2026. "Mechanisms, response heterogeneity, and dosing from MRI-derived electric field models in tDCS augmented cognitive training: a secondary data analysis of the ACT study**."**

G4. NSF NCS-FO-2318984, **PI, $1,158,999,** 2023-2027, "NCS-FO: Brain-Inspired Goal-Oriented and Bidirectional Deep Emotion Inference." + NSF INTERN Supplement + AI Supplement.

G5. NSF, IIS, **Co-PI, $840,000 (Fang: $301,176, Total $1.2M),** 2021-2026. "Collaborative Research: SCH: Trustworthy and Explainable AI for Neurodegenerative Diseases".

G6. RIEC Cooperative Research Projects, Tohoku University, **PI, 238,000 JPY**, 2026-2029. "Neurocomputational Principles Underlying Human Visual Abstraction. "

G7. NIH NIGMS, Co-I (PI: Azra Bihoric, Benjamin Shickel, Yulia Strekalova) Co-I, **$2,560,101,** 2024-2029 "R25: Artificial Intelligence Passport for Biomedical Research: Digital Experiential Learning Community for Upskilling in Artificial Intelligence for Biomedical, Behavioral and Clinical Research."

G8. NIH NIA, R01, **Site-PI** (PI: David Clark), **$4.98M**, 2023-2028. "R01: Cognitively engaging walking exercise and neuromodulation to enhance brain function in older adults".

G9. NSF, IIS-1908299, **PI, $823,000 (Fang: $796,619),** 2019-2024. "III: Small: Modeling Multi-Level Connectivity of Brain Dynamics" + REU Supplement.

G10. NSF, IIS-1564892, **Single PI, $190,991 (Fang: $190,991),** 2016-2018. "CRII: SCH: Characterizing, Modeling and Evaluating Brain Dynamics".

G11. ORAU, 800006901, **Single PI, $10,000 (Fang: $10,000),** 2016-2017. "Modeling, Estimating and Reasoning in Limited Data Brain Dynamics", Ralph Lowe Junior Faculty Enhancement Award, Oak Ridge Associated Universities (ORAU).

G12. Consortium for Medical Marijuana Clinical Outcomes, Co-I (PI: Adriaan Willem Bruijnzeel), **$75K (Fang: $5,749)**, 2025-2026. "Effects of Cannabis Constituents on the Reinforcing Properties of Fentanyl in Rats: A Behavioral and Neurobiological Approach" (only 6 funded)

G13. NIH NIAID, Co-I (MPI: Marco Salemi, Mattia Prosperi) **$2.8M (Fang: $86,475),** 2020-2025. "R01: Forecasting trajectories of HIV transmission networks with a novel phylodynamic and deep learning framework".

G14. NIH NIAID, P01AI165380, **Co-I** (PI: Gayle E. Woloschak, Northwestern University), **$10,703,529 ($1.95M to UF).** 2022-2027. "P01: Multi-Scale Evaluation and Mitigation of Toxicities Following Internal Radionuclide Contamination".

G15. NIH NIAAA, P01AA029543, **Co-I** (PI: Samuel Wu & Robert Cook), **$7,649,365 (Fang: $113,802).** 2021-2026. "P01: Interventions to improve alcohol-related comorbidities along the gut-brain axis in persons with HIV infection".

G16. NIH NIAAA, P01AA029547, **Co-I** (PI: Samuel Wu & Robert Cook), **$1,755,506 (Fang: $102,301)**. 2021-2026. "P01: SHARE Program: Innovations in Translational Behavioral Science to Improve Self-management of HIV and Alcohol Reaching Emerging adults."

G17. NIH NIAAA, U24AA029959, **Co-Investigator** (PI: Samuel Wu), **$2,491,414 (Fang: $70,185).** 2021-2026 "Southern HIV and Alcohol Research Consortium Biomedical Data Repository".

G18. NIH NIMH, R01MH125615, **Co-Investigator** (PI: Mingzhou Ding & Keil Andreas), **$2,279,906 (Fang: 115,817)**, 2021-2026. "Acquisition, extinction, and recall of attention biases to threat: Computational modeling and multimodal brain imaging".

G19. NIH NINDS, U01NS119562, **Co-Investigator** (PI: David Vaillancourt), **$5,264,645 (Fang: $69,973),** 2021-2026. "Web-based Automated Imaging Differentiation of Parkinsonism."

G20. Consortium for Medical Marijuana Clinical Outcomes**, Co-Investigator** (PI: Adriaan W. Bruijnzeel) **$75,000**, 7/1/2025-6/30/2026, "Effects of Cannabis Constituents on the Reinforcing Properties of Fentanyl in Rats: A Behavioral and Neurobiological Approach."

G21. NSF, CNS 1747783, **Co-Investigator** (PI: Jose Principe), **$900,478,** 2018-2022. "Phase I IUCRC University of Florida: Center for Big Learning"

G22. Department of Homeland Security (DHS), DHS-16-ST-062-001, **Co-Investigator** (PI: Jason Liu), **$1,200,000**, 2017-2020. "Scientific Leadership Awards for Minority Serving Institutions Granting Bachelor's Degrees"

G23. NSF, CNS-1560134, **Senior Personnel** (PI: Niki Pissinou), **$360,000**, 2016-2020. "REU SITE: ASSET: Research Experiences for Undergraduates in Advanced Secured Sensor Enabling Technologies"

G24. NSF, CNS-1263124 **Senior Personnel** (PI: Niki Pissinou), **$360,000,** 2013-2017, "REU SITE: ASSET: Research Experiences for Undergraduates in Advanced Secured Sensor Enabling Technologies"

G25. NSF, CNS-1407067, **Senior** Personnel (PI: Niki Pissinou), **$498,000**, 2014-2018. "RET in Engineering and Computer Science SITE: Research Experience for Teachers on Cyber-Enabled Technologies"

INTERNAL GRANTS

G26. UF BME Shepard Broad Foundation Community Outreach Fund, **PI**, $2000, Brain × AI 101: A Neuroscience-AI Curriculum Toolkit for High School AI Education, 2026-2027.

G27. UF Presidential Strategic Fund, **AI Lead**, UF StrokeInsight: AI-Powered Stroke Care, $1M, University of Florida.

G28. 2023 Research Opportunity Fund, **Co-PI, $86,500,** 2023-2025. "DRPD-ROF2023: Learning optimal treatment strategies for hypotension in critical care patients with acute kidney injury using artificial intelligence", University of Florida.

G29. Artificial Intelligence Research Catalyst Fund, **PI, $50,000 (Fang: $50,000),** 2020-2021. "VCA-DNN: Neuroscience-Inspired Artificial Intelligence for Visual Emotion Recognition", University of Florida,

G30. UFII-CTSI Pilot Award, **PI, $75,000 (Fang: $750,000**), 2018-2019. "Precision Dose: Personalized Radiation Dose Optimization for Multimodal Imaging", University of Florida.

G31. UFII Junior SEED Award, **Co-PI, $40,000 (PI: Wu; Fang: $20,000)**, 2019-2020. Multimodal Visual-Text Learning from Clinical Narrative and Image for Early Detection of Diabetic Retinopathy, University of Florida.

G32. UFII-CTSI Pilot Award, **Co-PI, $60,000 (PI: Gong; Fang: $20,000),** 2019-2020. Toward prevention of cardiotoxicity in cancer: a multimodal approach leveraging genomics, images and clinical data, University of Florida.

G33. Seed Grant, **Co-PI $5,552, (PI: Godavarty)**, 2016-2017. "Quantitative Differentiation Of Healing And Non-Healing Diabetic Ulcers Using Near-Infrared Optical Imaging", Florida International University.

G34. NIH CTSC, **Co-PI, $100,000, (PI: Gupta),** 2015-2017. "Minimal Radiation Exposure Technology For Acute Stroke Assessment In CT Perfusion Using Sparse Deconvolution And Dictionary Learning".

G35. Faculty Seed Grant for Collaborations Between Cornell University-Ithaca and Weill Cornell Medical College, **Co-Investigator $50,000**, **(PI: Chen, Gupta),** 2014-2015. Learning-Based Low Radiation CT Perfusion for Acute Stroke Diagnosis, Cornell University.

INDUSTRY SUPPORT

G36. NVIDIA Corporation**, PI,** Deep Learning GPU Equipment. 2016

G37. Oracle Research Award, **PI,** $100K, Cloud credit, 2022.

MENTEE GRANTS

G38. UF Informatics Institute, **Mentor (Mentee: Peng Liu), $11,616,** 2020-2021. "Biology and Cognition Inspired Deep Learning", University of Florida.

G39. NIH CTSI TL1 Predoctoral Fellowship. **Mentor (Mentee: Kyle B. See), $11,200,** 2019-2021. "Predicting short-term and long-term effects of spinal cord stimulation: implications for clinical practice" University of Florida

G40. NIH F31 Ruth L. Kirschstein Predoctoral Individual National Research Service Award, Co-sponsor (Mentee: Andrew Moore), $99,815, 2023-2025. "Identification of Prospective Predictors of Alcohol Initiation During Early Adolescence", National Institute of Health (NIH)/ National Institute on Alcohol Abuse and Alcoholism (NIAAA)

G41. NIH NIA K23 Using Artificial Intelligence to Predict Cognitive Training Response in Amnestic Mild Cognitive Impairment (Mentee: Joseph Gullet), 09/2022 - 08/2027, $833,352.

PUBLICATIONS

The authors with ∗ are my (co-)supervised students. Underline means I serve as the corresponding/senior author.

BOOK CHAPTERS

BC1. [Spring'24] Yanjun Li, Chaoyue Sun*, Daria Y. Romanova, Dapeng O. Wu, **Ruogu Fang**, Leonid L. Moroz. Chapter 17. Analysis and Visualization of Single-Cell Sequencing Data with Scanpy and MetaCell: A Tutorial, in Book

Ctenophores, Methods and Protocols, Springer, 2024. ISBN: 978-1-0716-3641-1 https://link.springer.com/book/9781071636411

BC2. [CRC'23] Yao Xiao*, Kai Huang, Hely Lin, **Ruogu Fang**: Medical Imaging Denoising, in Book Medical Image Synthesis, CRC Publisher, 2023 (invited). DOI: 10.1201/9781003243458-10

BC3. [Springer'19] Yao Xiao∗, Skylar Stolte∗, Peng Liu∗, Yun Liang∗, Pina Sanelli, Ajay Gupta, Jana Ivanidze, **Ruogu Fang**: Deep Spatial-Temporal Convolutional Neural Networks for Medical Image Restoration, in Book Deep Learning and Convolutional Neural Networks for Medical Image Computing, Springer Publisher, 2019.

BC4. [CRC'19] **Ruogu Fang**, Samira Pouyanfar, Yimin Yang, Yao Xiao∗, Jianqiao Tian∗, Shu-Ching Chen, S.S. Iyengar: Big Data in Computational Health Informatics, in Book Big Data in Multimodal Medical Imaging, CRC Publisher, 2019. (Invited)

PEER-REVIEWED JOURNAL

Highlight: Total 75 refereed journal articles, including 42 corresponding author journal papers. 8 first author. 2 Most Cited Journal Papers in MIA, 1 Nature BME, 1 in Nature Computational Science, 1 in Nature Communications, 1 in the Lancet Digital Health (IF=36.6), 1 in JAMA Neurology, 1 in npj Digital Medicine (IF=15.4), 1 in npj Parkinson's Disease, 1 in npj Women Health, 1 in npj Aging, 1 in PNAS Nexus, 10 in MIA (IF=13.8), 1 in IEEE TMI (IF=11.0), 1 in IEEE TCBY (IF=11.448), 1 in ACM CSUR (IF=14.3), 1 in RBME, 4 in Nature Scientific Reports, 2 in PLOS Computational Biology.

J1. [Nature BME'26] Sheng Liu, Long Chen, Zeyun Zhao*, Qinglin Gou, Qingyue Wei, Arjun Masurkar, Kevin M. Spiegler, Philip Kuball, Stefania C. Bray, Megan Bernath, Deanna R. Willis, Jiang Bian, Lei Xing, Eric Topol, Kyunghyun Cho, Yu Huang, **Ruogu Fang**, Narges Razavian, James Zou: A Multidisciplinary Agentic AI Board for Multimodal Dementia Characterization and Risk Assessment, Nature Biomedical Engineering, 2026. (accepted)

J2. [npj Aging'26] Wasif Khan*, Lin Gu, Noah Hammarlund, Lei Xing, Joshua K. Wong, **Ruogu Fang**: Association of Timing and Duration of Moderate-to-Vigorous Physical Activity with Cognitive Function and Brain Aging, Manuscript, 2026. (accepted)

J3. [Nature Comm'26] Alejandro Santos-Mayo, Faith Gilbert, Arash Mirifar, Anna-Lena Tebbe, **Ruogu Fang**, Mingzhou Ding, Andreas Keil: Concept2Brain: An AI model for predicting neurophysiological responses to text and pictures, Nat Commun 17, 8961 (2026). https://doi.org/10.1038/s41467-026-75653-x

J4. [NeuroImage'26] Tyler A. Busch, Skylar E. Stolte, Yunfeng Dai, Andrew O'Shea, Kevin Iversen, Samantha Pedersen, Gavin Hart, Jason Chen, Pratyush Shukla, Alejandro Albizu, **Ruogu Fang**, Adam J. Woods, Aprinda Indahlastari: Neuroanatomical predictors of frontal tDCS current distribution in cognitively healthy older adults, NeuroImage, 341:122243, 2026. https://doi.org/10.1016/j.neuroimage.2026.122243.

J5. [Emotion'26] Faith Gilbert, Fan Yang*, Ethan Smith*, Sarah Gardy, Andrew Farkas, Mingzhou Ding, **Ruogu Fang**, Andreas Keil: Emotional Responses to Naturalistic and AI-generated Affective Pictures: A Systematic Comparison, *Emotion*. 2026. Accepted.

J6. [NC'26] Seowung Leem*, Andreas Keil, Mingzhou Ding†, **Ruogu Fang**†: Associative Emotional Learning in Convolutional Neural Networks, Neural Computation. 2026. Accepted. (†: co-corresponding authors)

J7. [NN'26] Peng Liu*, Ke Bo, Yujun Chen, Andreas Keil, Mingzhou Ding†, **Ruogu Fang**†: Biologically Inspired Deep Neural Network Models for Visual Emotion Processing, Network Neuroscience, 2026. Accepted. (†: co-corresponding authors)

J8. [TBME'26] Yizheng Chen, Sheng Liu, Junyan Liu, Lequan Yu, **Ruogu Fang**, James Zou, Michael Francis Gensheimer, Lei Xing: A Single Reference-Guided Adaptation of Foundation Model Predictions for High-Performance Image Segmentation, IEEE Transaction on Biomedical Engineering. Accepted. 2026.

J9. [Frontiers'26] Junfu Cheng*, Skylar E. Stolte*, Zeyun Zhao*, Andrew O'Shea, Aprinda Indahlastari, Adam J. Woods, **Ruogu Fang**: Machine Learning and Individual Variability in Electrical Field Characteristics Predict tDCS Treatment Response for Anxiety in Older Adults in the ACT Trial, Frontiers in Human Neuroscience, 20, p.1847131, 2026. https://doi.org/10.3389/fnhum.2026.1847131

J10. [Frontiers'26] Wen Li, **Ruogu Fang**: Editorial: Women in Brain Imaging and Stimulation in the Era of Artificial Intelligence, Frontiers in Human Neuroscience, section Brain Imaging and Stimulation, 2026.

J11. [npjPD'26] Sneha Mantri, Rhoda Au, Anastasia Barnes, James Beck, Kathleen Blake, Wendy Dahl, Briana de Miranda, E. Ray Dorsey, **Ruogu Fang**, J. Daniel Fenn, Sue J. Goldie, Samuel M. Goldman, Paulina Gonzalez-Latapi, Diego Guarin, Brittany Kryzyzanowski, Nicole Lessard, Andi Lipstein Fristedt, Tejas R. Mehta, Monica Norcini, Peggy Borum, Jennifer L. Purks, Dawn Schiehser, Michael A. Schwarzschild, Caroline M. Tanner, Alexis Temkin, J. Bryan Unruh, Chris Vulpe, Kristin Wallock, Cordelia Walz, Michael S. Okun: Primary Prevention of Parkinson's Disease: Proceedings, the 7 C's and a Position Statement from the Parkinson's Disease (PD) Prevention Think Tank, *npj Parkinson's Disease*, 2026.

J12. [JAD'26'] Seowung Leem*, Yunchao Yang, Adam J. Woods, **Ruogu Fang**. Prediction of Alzheimer's Disease Risk Factors from Retinal Images via Deep Learning: Development and Validation of Biologically Relevant Morphological Associations in the UK Biobank. Journal of Alzheimer's Disease. June 2026. p.13872877261457650. https://doi.org/10.1177/13872877261457650

J13. [COBE'26] **Ruogu Fang**, Parisa Rashidi: Editorial: Artificial Intelligence in Biomedical Engineering, Current Opinion in Biomedical Engineering, March 16, 2026, Article 100661. https://doi.org/10.1016/j.cobme.2026.100661

J14. [TAI'26] Junfu Cheng*, Tara Sahni*, Zeyun Zhao*, Skylar E. Stolte*, Chenyu You, Adam J. Woods, Aprinda Indahlastari, **Ruogu Fang**. Recent Advancements of Transcranial Direct Current Stimulation and Machine Learning: Methods, Challenges, and Opportunities. Transactions on Artificial Intelligence, 2(1), 54-77, 2026. https://doi.org/10.53941/tai.2026.100005

J15. [Kidney360'26] Esra Adiyeke, Tianqi Liu, Venkata Sai Dheeraj Naganaboina, Han Li, Tyler J. Loftus, Yuanfang Ren, Benjamin Shickel, Matthew M. Ruppert, Karandeep Singh, **Ruogu Fang**, Parisa Rashidi, Azra Bihorac, Tezcan Ozrazgat-Baslanti. Learning Optimal Treatment Strategies for Intraoperative Hypotension Using Deep Reinforcement Learning. Kidney360, accepted.

J16. [Frontier'26] Wasif Khan*, John Rees, Kyle B. See*, Simon Kato*, Ziqian Huang*, Amy Lazarte*, Kyle Douglas*, Xiangyang Lou, Teng J. Peng, Dhanashree Rajderkar, Pina Sanelli, Amita Singh, Ibrahim Tuna, Christina A. Wilson, **Ruogu Fang**: Diagnostically competitive performance of a physiology-informed generative multi-task network for contrast-free CT perfusion, Frontiers in Human Neuroscience, 2026. https://doi.org/10.3389/fnhum.2026.1703004

J17. [SN'25] Tejas R. Mehta, Venkat S. Lavu, Hao Gao, Tania Banerjee, Renjie Hu, **Ruogu Fang**, Joshua K. Wong: The Role of Artificial Intelligence in Deep Brain Stimulation, Seminars in Neurology, 2026. DOI: 10.1055/a-2753-6166

J18. [NatureCS'25] Zixia Zhou, Junyan Liu, Wei Emma Wu, **Ruogu Fang**, Sheng Liu, Qingyue Wei, Rui Yan, Yi Guo, Qian Tao, Yuanyuan Wang, Md Tauhidul Islam, Lei Xing: Revealing Neurocognitive and Behavioral Patterns by Self-Supervised Manifold Learning from Dynamic Brain Data, *Nature Computational Science* (IF=18), 2025. https://doi.org/10.1038/s43588-025-00911-9

J19. [NPJWH'25] Diandra P. Ojo*, Cameron Celeste, Dion Ming, **Ruogu Fang**†, Ivana K. Parker†: Population-Level Predictive Variation in Machine Learning Diagnosis of Symptomatic Bacterial Vaginosis, *npj Women's Health*, 2025. (†: co-corresponding authors) https://doi.org/10.1038/s44294-025-00092-w

J20. [RBME'25] Wasif Khan*, Seowung Leem*, Kyle B. See*, Joshua K. Wong, Shaoting Zhang, **Ruogu Fang**. A comprehensive survey of foundation models in medicine. IEEE Reviews in Biomedical Engineering, 2025. https://doi.org/10.1109/RBME.2025.3531360

J21. [UFUG'25] Silva, Giovana, Tyler Busch, Samantha Pedersen, Kevin Iversen, Priya Sambasivan, Ruogu Fang, Adam Woods, Aprinda Indahlastari Queen: The Importance of Tissue Identity in Modeling Transcranial Direct Current Stimulation, UF Journal of Undergraduate Research 27, 2025. https://doi.org/10.32473/ufjur.27.138665

J22. [FrontierDH'25] Alison M. Veintimilla#, Chintan K. Acharya#*, Connie J. Mulligan, **Ruogu Fang**†, Erika Moore†: TRACE: An AI-Powered Tool for Ancestry Meta-Analysis to Promote Transparency and Representation in Biomedical Research, Frontiers in Digital Health, 2025 (#: co-first authors, ,†: co-corresponding)

J23. [FrontierHN'25] Editorial: Frontiers of Women in Brain Imaging and Stimulation, Volume II, Molly Suzanne Hermiller, **Ruogu Fang**, Wen Li, Editorial, Frontiers in Human Neuroscience—Brain Imaging and Stimulation. https://doi.org/10.3389/fnhum.2025.1648769

J24. [JAMA'25] David Vaillancourt, Angelos Barmpoutis, Samuel Wu, Jesse DeSimone, Marissa Schauder, Robin Chen, Todd Parrish, Wei-en Wang, Eric Molho, Alicia Leader, John Morgan, Áine Russell, David Simon, Hannah Babcock, Karen White-Tong, Burton Scott, Jun Hua, Liana Rosenthal, Stephen Gomperts, Anna Goodheart, Erin Peterec, Cynthia Poon, Max Galarce, Rizwan Akhtar, Tanya Thompson, David Grimes, Sol De Jesus, Autumn Collier, Candace Cromer, Natividad Stover, Ece Bayram, Natt Putra, Reilly Costello, Eda Yilmaz, Crystal Mercado, Tomas Mercado, Adolfo Ramirez-Zamora, Amanda Fessenden, Stefan Prokop, **Ruogu Fang**, Renee Wagner, John Slevin, C Spears, Prabesh Kanel, Nicolaas Bohnen, Jacqueline Caswell, Marina Bryants, Paul Tuite, Kristyn Kuzianik, Youshra Ahmed, Stephen Aradi, Antonio Strafella, Nathaniel Bendahan, Mustafa Siddiqui, Joy Njoku, Albert Davis, Amy Stiebel, Xuemei Huang, Jill Ostrem, Hengameh Zahed, Sarah Wang, Phuong Hoang, Joseph Seemiller, Guangwei Du, Hubert Fernandez, Irene Litvan, Robert Hauser, Alexander Pantelyat, Nikolaus McFarland, Tao Xie, Michael Okun: Prospective Multicenter Study of Automated Imaging Differentiation for Parkinsonism (AIDP), JAMA Neurology, 2025. doi:10.1001/jamaneurol.2025.0112

J25. [BA'24] Chaoyue Sun*, Yanjun Li, Simone Marini, Alberto Riva, Dapeng Oliver Wu, **Ruogu Fang**, Marco Salemi, and Brittany Rife Magalis: Phylogenetic-informed graph deep learning to classify dynamic transmission clusters in infectious disease epidemics, in *Bioinformatics Advances* 4, no. 1, vbae158, 2024. https://doi.org/10.1093/bioadv/vbae158

J26. [PNAS'24] Vadim N. Gladyshev, … **Ruogu Fang**, et al., Disagreement on foundational principles of biological aging, *Proceedings of the National Academy of Sciences* (*PNAS*) *Nexus*, 2024. https://doi.org/10.1093/pnasnexus/pgae499

J27. [BM'24] Alejandro Albizu, Paulo Suen, Ziqian Huang*, Jori L. Waner, Skylar E. Stolte*, Aprinda Indahlastari, **Ruogu Fang**, Andre R. Brunoni, and Adam J. Woods: Artificial Intelligence-Optimized Non-Invasive Brain Stimulation and Treatment Response Prediction for Major Depression. Bioelectronic Medicine, 2024. https://doi.org/10.1186/s42234-024-00157-2

J28. [MIA'24] Joseph Cox*, Peng Liu*, Skylar E. Stolte*, Yunchao Yang, Kang Liu*, Kyle B. See*, Huiwen Ju, **Ruogu Fang**: BrainSegFounder: Towards 3D Foundation Models for Neuroimage Segmentation, Medical Image Analysis, 2024. https://doi.org/10.1016/j.media.2024.103301

J29. [BB'2024] Qiaosi Tang, Ranjala Ratnayake, Gustavo Seabra, Zhe Jiang, **Ruogu Fang,** Lina Cui, Yousong Ding, Tamer Kahveci, Jiang Bian, Chenglong Li, Hendrik Luesch, Yanjun Li: Morphological Profiling for Drug Discovery in the Era of Deep Learning, Briefings in Bioinformatics, accepted. (Impact factor=9.5)

J30. [PLOS-CB'24] Chaoyue Sun*, **Ruogu Fang**, Marco Salemi, Brittany Rife Magalis, Mattia Prosperi: DeepDynaForecast: Phylogenetic-informed graph deep learning for epidemic transmission dynamic prediction. PLOS Computational Biology, 20(4), p.e1011351, 2024. https://doi.org/10.1371/journal.pcbi.1011351

J31. [Nature SR'24] Nooshin Hosseini Yousefzadeh, Charlie Train*, **Ruogu Fang**, My Thai: LAVA: Granular Neuron-Level Explainable AI for Alzheimer's Disease Assessment from Fundus Images, Nature Scientific Reports, https://doi.org/10.1038/s41598-024-58121-8

J32. [PLOS CB'24] Peng Liu*, Ke Bo, Mingzhou Ding, **Ruogu Fang**: Emergence of Emotion Selectivity in A Deep Neural Network Trained to Recognize Visual Objects. PLOS Computational Biology, 2024. https://doi.org/10.1371/journal.pcbi.1011943

J33. [Nature SR'24] Charlie Tran*, Kai Shen*, Kang Liu*, Akshay Ashok*, Adolfo Ramirez-Zamora4 Jinghua Chen, Yulin Li, and **Ruogu Fang**: Explainable Deep Learning Predicts Prevalent and Incident Subjects of Parkinson's Disease from UK Biobank Fundus Imaging. Nature Scientific Reports, 2024. https://doi.org/10.1038/s41598-024-54251-1

J34. [IN'24] Skylar E. Stolte*, Aprinda Indahlastari, Jason Chen*, Alejandro Albizu, Ayden Dunn, Samantha Pedersen, Kyle B. See*, Adam J. Woods, **Ruogu Fang**: Precise and Rapid Whole-Head Segmentation from Magnetic Resonance Images of Older Adults using Deep Learning, Imaging Neuroscience, 2024. https://doi.org/10.1162/imag_a_00090

J35. [npj Digital Medicine'23] Cameron Celeste, Dion Ming, Justin BroceU, Diandra P. Ojo, Emma Drobina, Adetola F. Louis-Jacques, Juan E. Gilbert, **Ruogu Fang**†, Ivana K. Parker†. Disparity in Diagnosing Bacterial Vaginosis using Machine Learning. *npj Digital Medicine* 6:211, 2023, collection on Women's Health and Maternal Care. († co-corresponding authors) (**Featured by UF News, EurekAlert, and Newswise**) https://doi.org/10.1038/s41746-023-00953-1

J36. [JVCIR'24] Tianqi Liu*, Hong Huang*, Zhijun Lei, **Ruogu Fang**, Dapeng Wu: Texture and motion aware perception in-loop filter for AV1, Journal of Visual Communication and Image Representation, vol. 98, 104025. https://doi.org/10.1016/j.jvcir.2023.104025

J37. [BS'23] Alejandro Albizu, Aprinda Indahlastari, Ziqian Huang*, Jori Waner, Skylar E. Stolte*, **Ruogu Fang**†, Adam J. Woods†: Machine-Learning Defined Precision tDCS for Improving Cognitive Function. Brain Stimulation, June 4, 2023. DOI: https://doi.org/10.1016/j.brs.2023.05.020 PMID: 37279860 (†: co-corresponding authors)

J38. [Frontiers'23] **Ruogu Fang**, Lijun Bai, Wen Li. Editorial: Frontiers of women in brain imaging and brain stimulation, Frontiers in Human Neuroscience, 17:1208253, 2023. http://doi.org/10.3389/fnhum.2023.1208253

J39. [AA'22] Adam J. Woods, Gene Alexander, Ronald Cohen, Michael Marsiske, Steven T DeKosky, Georg A. Hishaw, Samuel S. Wu, Aprinda Indahlastari, Alejandro Albizu, **Ruogu Fang:** Augmenting cognitive training in older adults with transcranial direct current stimulation: Initial results from the Phase III ACT trial and new directions, Alzheimer's Association, December 2022.

J40. [SI'22] Skylar Stolte*, Kyle Volle, Aprinda Indahlastari, Alejandro Albizu, Adam Woods, Kevin Brink, Matthew Hale, **Ruogu Fang**: DOMINO: Domain-aware Loss for Deep Learning Calibration, Software Impacts, p.100478, 2023. https://doi.org/10.1016/j.simpa.2023.100478

J41. [NoA'22] Wei-en Wang, Rob Chen, Robin Perry Mayrand, Malek Adjouadi, **Ruogu Fang**, Steven T. DeKosky, Ranjan Duara, Stephen A. Coombes, David E. Vaillancourt*, for the Alzheimer's Disease Neuroimaging Initiative: Association of Longitudinal Cognitive Decline with Diffusion Changes in Gray Matter, and Amyloid and Tau Deposition, in Neurobiology of Aging, 121, pp.166-178, 2023. https://doi.org/10.1016/j.neurobiolaging.2022.10.013

J42. [BMC'22] Zehao Yu, Xi Yang, Gianna L. Sweeting*, Yinghan Ma, Skylar E. Stolte*, **Ruogu Fang**, Yonghui Wu: Identify diabetic retinopathy-related clinical concepts and their attributes using transformer-based natural language processing methods, in BMC Medical Informatics and Decision Making, Sep. 2022. https://doi.org/10.1186/s12911-022-01996-2

J43. [Frontiers'22] Peng Liu*, Linsong Xu, Garrett Fullerton*, Yao Xiao*, James-Bond Nguyen, Zhongyu Li, Izabella Barreto, Catherine Olguin, **Ruogu Fang**: PIMA-CT: Physical Model-Aware Cyclic Simulation and Denoising for Ultra-low-dose CT Restoration, in *Frontiers in Radiology*, section Artificial Intelligence in Radiology, May 25, 2022. https://doi.org/10.3389/fradi.2022.904601

J44. [NeuroImage'21] Kyle B. See*, David J. Arpin, David E. Vaillancourt, **Ruogu Fang**†, Stephen A. Coombes†: Unraveling Somatotopic Organization in the Human Brain using Machine Learning and Adaptive Supervoxel-based Parcellations, in *NeuroImage*, vol. 245, 118710, 16 pages, 2021. († co-corresponding authors) https://doi.org/10.1016/j.neuroimage.2021.118710

J45. [NC'21] Peng Liu*, Charlie T. Tran*, Bin Kong, **Ruogu Fang**, CADA: Multi-scale Collaborative Adversarial Domain Adaptation for Unsupervised Optic Disc and Cup Segmentation, in *NeuroComputing*, vol. 469, pp. 209-220, 2021. https://doi.org/10.1016/j.neucom.2021.10.076

J46. [Frontiers'21] Joseph M. Gullett, Alejandro Albizu, **Ruogu Fang**, David A. Loewenstein, Ranjan Duara, Monica Rosselli, Melissa J. Armstrong, Tatjana Rundek, Hanna K. Hausman, Steven T. Dekosky, Adam J. Woods, Ronald A. Cohen. Baseline neuroimaging predicts decline to dementia from amnestic mild cognitive impairment. Frontiers in Aging Neuroscience, 21 pages, 2021. (accepted). **Featured by UF Health**

J47. [TCFD'21] Siddani, Bhargav, S. Balachandar, William C. Moore, Yunchao Yang, **Ruogu Fang**. Machine learning for physics-informed generation of dispersed multiphase flow using generative adversarial networks, in *Theoretical and Computational Fluid Dynamics*, 24 pages, 2021. https://doi.org/10.1007/s00162-021-00593-9

J48. [PoF'21] Bhargav Siddani, S. Balachandar, and **Ruogu Fang**. Rotational and reflectional equivariant convolutional neural network for data-limited applications: Multiphase flow demonstration, in *Physics of Fluids,* vol. 33, no. 10, Article ID 103323, 18 pages, 2021. https://doi.org/10.1063/5.0066049 (Editor's Pick)

J49. [Nature SR'21] Jianqiao Tian*, Glenn Smith, Han Guo, Boya Liu, Zehua Pan, Zijie Wang, Shuangyu Xiong, **Ruogu Fang**: Modular machine learning for Alzheimer's disease classification from retinal vasculature, in *Nature Scientific Reports*, vol. 11, no. 1, Article ID 238, 11 pages, 2021. https://doi.org/10.1038/s41598-020-80312-2. **Highlighted in The Washington Post, UF Engineering News.**

J50. [JCTS'21] Kyle See*, Rachel Ho, Steven Coombes, **Ruogu Fang**: TL1 Team Approach to Predicting Response to Spinal Cord Stimulation for Chronic Low Back Pain, in *Journal of Clinical and Translational Science*, vol. 5, no. s1, pp. 111-112, 2021. https://doi.org/10.1017/cts.2021.685

J51. [MIA'20] Skylar E. Stolte*, **Ruogu Fang**: A Survey on Medical Image Analysis in Diabetic Retinopathy, in *Medical Image Analysis*, vol. 64, Article ID 101742, 27 pages, 2020. https://doi.org/10.1016/j.media.2020.101742 **Impact Factor: 8.545**

J52. [JCTS'20] Kyle See*, Rachel Louise Mahealani Judy*, Stephen Coombes, **Ruogu Fang**: TL1 Team Approach to Predicting Short-term and Long-term Effects of Spinal Cord Stimulation, in *Journal of Clinical and Translational Science*, vol. 4, no. s1, pp. 120, 2020. https://doi.org/10.1017/cts.2020.362

J53. [BrainStim'20] Alejandro Albizu, **Ruogu Fang**, Aprinda Indahlastari, Andrew OShea, Skylar E. Stolte*, Kyle B. See*, Emanuel M. Boutzoukas, Jessica N. Kraft, Nicole R. Nissim, and Adam J. Woods: Machine learning and individual variability in electric field characteristics predict tDCS treatment response. in *Brain Simulation*, vol. 13, no. 6, 2020, pp. 1753-1764. **Impact Factor: 8.955** https://doi.org/10.1016/j.brs.2020.10.001 **Featured by Ivanhoe, Alligator, UF Health.**

J54. [JOSAA'20] E. A. Robledo, R. Schutzman, **R. Fang**, C. Fernandez, R. Kwasinski, K. Leiva, F. Perez-Clavijo, A. Godavarty: Physiological wound assessment from coregistered and segmented tissue hemoglobin maps, in *Journal of the Optical Society of America A*, vol. 37, no. 8, pp. 1249-1256, 2020. https://doi.org/10.1364/JOSAA.394985

J55. [MIA'20] Yaxin Shen, Bin Sheng, **Ruogu Fang**, Huating Li, Ling Dai, Skylar Stolte*, Jing Qin, Weiping Jia, Dinggang Shen: Domain-Invariant Interpretable Fundus Image Quality Assessment, in *Medical Image Analysis,* vol. 61, Article ID 101654, 15 pages, 2020. https://doi.org/10.1016/j.media.2020.101654 **Impact Factor: 8.545. (co-corresponding author)**

J56. [JWEIA'20] Jianqiao Tian*, Kurtis Gurley, Maximillian Diaz, Pedro L. Fernandez Caban, Forrest J. Masters, **Ruogu Fang**: Low-Rise Gable Roof Buildings Pressure Prediction using Deep Neural Networks, in *Journal of Wind Engineering Industrial Aerodynamics*, vol. 196, Article ID 104026, 17 pages, 2020. https://doi.org/10.1016/j.jweia.2019.104026

J57. [MIA'20] Prasanna Porwal, Samiksha Pachade, Manesh Kokare, Girish Deshmukh, Jaemin Son, Woong Bae, Lihong Liu, Jianzong Wang, Xinhui Liu, Liangxin Gao, TianBo Wu, Jing Xiao, Fengyan Wang, Baocai Yin, Yunzhi Wang, Gopichandh Danala, Linsheng He, Yoon Ho Choi, Yeong Chan Lee, Sang-Hyuk Jung, Zhongyu Li, Xiaodan Sui, Junyan Wu, Xiaolong Li, Ting Zhou, Janos Toth, Agnes Baran, Avinash Kori, Sai Saketh Chennamsetty, Mohammed Safwan, Varghese Alex, Xingzheng Lyu, Li Cheng, Qinhao Chu, Pengcheng Li, Xin Ji, Sanyuan Zhang, Yaxin Shen, Ling Dai, Oindrila Saha, Rachana Sathish, Tania Melo, Teresa Araujo, Balazs Harangi, Bin Sheng, **Ruogu Fang**, Debdoot Sheet, Andras Hajdu, Yuanjie Zheng, Ana Maria Mendonca, Shaoting Zhang, Aurelio Campilho, Bin Zheng, Dinggang Shen, Luca Giancardo, Gwenole Quellec, Fabrice Meriaudeau: IDRiD: Diabetic Retinopathy Segmentation and Grading Challenge, in *Medical Image Analysis*, vol. 59, Article ID 101561, 26 pages, 2020. https://www.sciencedirect.com/science/article/pii/S1361841519301033. **Impact Factor: 8.545. Our team won First Place in the International Diabetic Retinopathy Grading and Segmentation Challenge**

J58. [MIA'20] Jose Ignacio Orlando, Huazhu Fu, Joao Barbossa Breda, Karel van Keer, Deepti R. Bathula, Andres Diaz-Pinto, **Ruogu Fang**, Pheng-Ann Heng, Jeyoung Kim, JoonHo Lee, Joonseok Lee, Xiaoxiao Li, Peng Liu*, Shuai Lu, Balamurali Murugesan, Valery Naranjo, Sai Samarth R. Phaye, Sharath M. Shankaranarayana, Apoorva Sikka,

Jaemin Son, Anton van den Hengel, Shujun Wang, Junyan Wu, Zifeng Wu, Guanghui Xu, Yongli Xu, Pengshuai Yin, Fei Li, Xiulan Zhang, Yanwu Xu, Hrvoje Bogunovic : REFUGE Challenge: A unified framework for evaluating automated methods for glaucoma assessment from fundus photographs, in *Medical Image Analysis*, vol. 59, Article ID 101570, 21 pages, 2020. https://doi.org/10.1016/j.media.2019.101570. **Impact Factor: 8.545** **Top 25 Most Cited Paper since 2018**

J59. [Lancet'19] Derek B Archer, Justin T Bricker, Winston T Chu, Roxana G Burciu, Johanna L Mccracken, Song Lai, Stephen A Coombes, **Ruogu Fang**, Angelos Barmpoutis, Daniel M Corcos, Ajay S Kurani, Trina Mitchell, Mieniecia L Black, Ellen Herschel, Tanya Simuni, Todd B Parrish, Cynthia Comella, Tao Xie, Klaus Seppi, Nicolaas I Bohnen, Martijn L T M Müller, Roger L Albin, Florian Krismer, Guangwei Du 15, Mechelle M Lewis, Xuemei Huang, Hong Li, Ofer Pasternak, Nikolaus R McFarland, Michael S Okun, David E Vaillancourt. Vaillancourt: Development and Validation of the Automated Imaging Differentiation in Parkinsonism (AID-P): A Multi-Site Machine Learning Study, in *The Lancet Digital Health*, vol. 1, no. 5, pp. e222-e231, 2019. https://doi.org/10. 1016/S2589-7500(19)30105-0 **Featured on the Cover of Lancet Digital Health**

J60. [JAMIA'19] Xi Yang, Jiang Bian, **Ruogu Fang**, Ragnhildur I. Bjarnadottir, William R. Hogan, Yonghui Wu: Identifying Relations of Medications with Adverse Drug Events Using Recurrent Convolutional Neural Networks and Gradient Boosting, in *Journal of the American Medical Informatics Association*, 2019. https://doi.org/10.1093/jamia/ocz144. **Impact Factor: 4.112**

J61. [MIA'19] Peng Liu*, Yangjunyi Li*, Mohammad D EI Basha*, Yao Xiao*, Pina C. Sanelli, **Ruogu Fang**: Deep Evolutionary Networks with Expedited Genetic Algorithm for Medical Image Denoising, in *Medical Image Analysis*, vol. 54, pp. 306-315, 2019. https://doi.org/10.1016/j.media.2019.03.004 **Impact Factor: 13.8**

J62. [Frontiers'19] Yao Xiao*, Peng Liu*, Yun Liang, Skylar Stolte*, Pina Sanelli, Ajay Gupta, Jana Ivanidze, **Ruogu Fang**: STIR-Net: Deep Spatial-Temporal Image Restoration Net for Radiation Reduction in CT Perfusion, in *Frontiers in Neurology*, vol. 10, Article ID 647, 16 pages, 2019. https://doi.org/10.3389/fneur.2019.00647 **Impact Factor: 4.0**

J63. [Nature SR'19] Saleha Masood*, **Ruogu Fang**, Huating Li, Bin Sheng, Ping Li, Akash Mathavan, Xiangning Wang, Po Yang, Qiang Wu, Jing Qin, Weiping Jia: Automatic Choroid Layer Segmentation from Optical Coherence Tomography Images Using Deep Learning, in *Nature Scientific Reports*, vol. 9, no. 1, Article ID 3058, 18 pages, 2019. https://doi.org/10.1038/s41598-019-39795-x

J64. [CIN'18] Maryamossadat Aghili*, **Ruogu Fang**: Mining Big Neuron Morphological Data, in *Computational Intelligence and Neuroscience,* Article ID 8234734, 13 pages, 2018. https://doi.org/10.1155/2018/8234734.

J65. [TCBY'18] Bin Sheng, Ping Li, Shuangjia Mo, Huating Li, Xuhong Hou, Qiang Wu, Jing Qin, **Ruogu Fang**, David Dagan Feng: Retinal Vessel Segmentation Using Minimum Spanning Superpixel Tree Detector, in *IEEE Transaction on Cybernetics,* Article ID 99, 13 pages, 2018. **Impact Factor: 11.448** 10.1109/TCYB.2018.2833963

J66. [IET'17] Saleha Masood*, Bin Sheng, Ping Li, Ruimin Shen, **Ruogu Fang**, Qiang Wu: Automatic Choroid Layer Segmentation Using Normalized Graph Cut, in *IET Image Processing*, vol. 12, no. 1, pp. 53-59, 2017. https://doi.org/10.1049/iet-ipr.2017.0273

J67. [PR'17] Zhongyu Li, **Ruogu Fang**, Fumin Shen, Shaoting Zhang: Indexing and Mining Large-Scale Neuron Databases using Maximum Inner Product Search, in *Pattern Recognition*, vol. 63, no. C, pp. 680-688, 2017. https://doi.org/10.1016/j.patcog.2016.09.041

J68. [NC'17] Fei Jiang*, Huating Li, Xuhong Hou, Bin Sheng, Ruimin Shen, Xiao-Yang Liu, Weiping Jia, Ping Li, **Ruogu Fang**: Abdominal Adipose Tissues Extraction Using Multi-Scale Deep Neural Network, in *Neurocomputing*, vol. 229, no. C, pp. 23-33, 2017. **Impact Factor: 4.438** https://doi.org/10.1016/j.neucom.2016.07.059

J69. [NC'17] **Ruogu Fang**, Ajay Gupta, Junzhou Huang, Pina Sanelli: TENDER: TEnsor Non-local Deconvolution Enabled Radiation Reduction in CT Perfusion, in *Neurocomputing*, vol. 229, pp. 13-22, 2017. https://doi.org/10.1016/j.neucom.2016.03.109 **Impact Factor: 4.438**

J70. [CSUR'16] **Ruogu Fang**, Samira Pouyanfar*, Yimin Yang, Shu-Ching Chen, and S. S. Iyengar: Computational Health Informatics in the Big Data Age: A Survey, in *ACM Computing Survey*, vol. 49, no. 1, pp.1-36, 2016. https://doi.org/10.1145/2932707 **Impact Factor: 7.99**

J71. [TMI'15] **Ruogu Fang**, Shaoting Zhang, Tsuhan Chen, Pina C. Sanelli: Robust Low-dose CT Perfusion Deconvolution via Tensor Total-Variation Regularization, in *IEEE Transaction on Medical Imaging*, vol. 34, no. 7, pp.1533-1548, 2015. **Impact Factor: 10.048** https://doi.org/10.1016/j.compmedimag.2015.04.008

J72. [CMIG'15] **Ruogu Fang**, Tsuhan Chen, Dimitris Metaxas, Pina Sanelli, Shaoting Zhang: Guest Editorial: Sparsity Techniques in Medical Imaging, in *Computerized Medical Imaging and Graphics*, vol. 46, no. 1,p.1, 2015. https://doi.org/10.1016/j.compmedimag.2015.07.002

J73. [CMIG'15] **Ruogu Fang**, Haodi Jiang*, Junzhou Huang: Tissue-Specific Sparse Deconvolution for Brain CT Perfusion, in *Computerized Medical Imaging and Graphics*, vol. 46, no. 1, pp. 64-72, 2015. https://doi.org/10.1016/j.compmedimag.2015.04.008

J74. [MIA'14] **Ruogu Fang**, Kolbeinn Karlsson*, Tsuhan Chen, Pina C. Sanelli: Improving Low-Dose Blood-Brain Barrier Permeability Quantification Using Sparse High-Dose Induced Prior for Patlak Model, in *Medical Image Analysis*, 18(6), pp. 866-880, 2014. https://doi.org/10.1016/j.media.2013.09.008 **Impact Factor: 10.9 (2024)**

J75. [MIA'13] **Ruogu Fang**, Tsuhan Chen, Pina C. Sanelli: Towards Robust Deconvolution of Low-Dose Perfusion CT: Sparse Perfusion Deconvolution Using Online Dictionary Learning, in *Medical Image Analysis*, 17(4), pp. 417-428, 2013. https://doi.org/10.1016/j.media.2013.02.005 **Impact Factor: 10.9 (2024). Top 25 Hottest Articles in Medical Image Analysis in 2013 April-June**

PEER-REVIEWED CONFERENCE/WORKSHOP PUBLICATIONS

Highlight: Total 118 conference publications, including 38 referred proceedings, 17 first author papers and 82 senior author papers (underlined), 15 in MICCAI (leading conference in Medical Image Analysis), 1 ICCV, 3 RSNA, 2 Best Paper Awards, 1 Best Abstract Award, 1 Presential Trainee Award, 1 Early Career Scientist Award, 1 NextGen Scholar Award, 4 Travel Awards, 1 NSF REU Best Poster Award.

C1. [MICCAI'26] Ethan Meidinger*, Seowung Leem*, Zeyun Zhao*, **Ruogu Fang**: REVEAL++: Differentiable Phenotypic Grouping for Vision–Language Retinal Modeling of Alzheimer's Disease Risk, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Strasbourg, France, September 2026. (Acceptance Rate = 28%, 10 pages)

C2. [AD/PD'27] Qinglin Gou, Zeyun Zhao, Xuguang Ai, **Ruogu Fang**, Jiang Bian, Yu Huang: An Agentic Multimodal Digital Care Team for Dementia Diagnosis, International Conference on Alzheimer's and Parkinson's Diseases and Related Neurological Disorders (AD/PD), Barcelona, Spain, March 16-20, 2027.

C3. [BMES'26] Ethan Elio Meidinger*, Seowung Leem, **Ruogu Fang**. Multi-Modal Contrastive Representation Learning for Preclinical AD Prediction. Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL, October 21-24, 2026.

C4. [BMES'26] Zeyun Zhao*, Junfu Cheng*, Kimberly Sibille, Samuel Wu, Roland Staud, Angela Mickle, Selenia Rubio, Robert Sorge, Kevin Riggs, Fadel Zeidan, Hyochol Ahn, Adam Woods, Jared Tanner, Song Lai, Burel Goodin, Roger Fillingim, **Ruogu Fang**. Baseline Neuroimaging and Machine Learning Predict Pain Relief Following BAT-Paired tDCS in Knee Osteoarthritis. Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL, October 21-24, 2026.

C5. [BMES'26] Junfu Cheng*, Zeyun Zhao*, Andrew O'Shea, Aprinda Indahlastari, Adam Woods, **Ruogu Fang**. Clinico-Demographic Predictors of Anxiety Treatment Response to Transcranial Direct Current Stimulation and Cognitive Training in Older Adults. Biomedical Engineering Society (BMES) Annual Meeting, 2026.

C6. [BMES'26] Tara Sahni*, August Zhang*, Junfu Cheng*, **Ruogu Fang**. A Multimodal Vision-Language Model Expert Agent for Symptom-Informed Precision tDCS Dosing in Older Adults With Anxiety. Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL, October 21-24, 2026.

C7. [BMES'26] Peter Kowalski, **Ruogu Fang**, Hayden Good, Carlos M Rinaldi-Ramos. Improving Iron Mass Estimation in Magnetic Particle Imaging Using Deep Learning Modeling. Biomedical Engineering Society (BMES) Annual Meeting, Orlando, FL, October 21-24, 2026.

C8. [NYCN'26] Chintan K. Acharya*, Dev Rakesh Patel*, Skylar E. Stolte*, Aprinda Indahlastari, **Ruogu Fang**: Comprehensive Reconstruction Of Whole-head Neuromodulation (CROWN): A Browser-Based Platform for Automated 11-Tissue MRI Segmentation and Individualized tDCS Current Density Modeling. NYC Neuromodulation Conference, New York, NY, USA, June 5–6, 2026.

C9. [NYCN'26] Junfu Cheng*, Skylar E. Stolte*, Zeyun Zhao*, Andrew O'Shea, Aprinda Indahlastari, Adam J. Woods, **Ruogu Fang**: Machine Learning Prediction of Anxiety Response to Combined tDCS and Cognitive Training Using MRI-Derived Current Density Modeling in Older Adults, NYC Neuromodulation Conference, 2026

C10. [NYCN'26] Zeyun Zhao*, Junfu Cheng*, Roger Fillingim, Burel Goodin, Kim Sibille, Sam Wu, Selenia Rubio, Roland Staud, Robert Sorge, Kevin Riggs, Angela Mickle, Fadel Zeidan, Hyochol Ahn, Adam Woods, Jared Tanner, Song Lai, **Ruogu Fang**: Predicting Response to Combined BAT and tDCS in Knee Osteoarthritis Using Machine Learning and Baseline Neuroimaging, NYC Neuromodulation Conference, 2026

C11. [SNMMI'26] Boxiao Yu, Savas Ozdemir, Jiong Wu, Yizhou Chen, **Ruogu Fang**, Tinsu Pan, Axel Rominger, Kuangyu Shi, Kuang Gong. Dose- and Anatomy-Adaptive Whole-Body PET Image Denoising Using a Vision-Language Model and Diffusion Model. SNMMI, 2026.

C12. [MIDL'26] Seowung Leem*, Lin Gu, Chenyu You, Kuang Gong, **Ruogu Fang**: REVEAL: Multimodal Vision–Language Alignment of Retinal Morphometry and Clinical Risks for Incident AD and Dementia Prediction, Medical Imaging with Deep Learning (MIDL), Jul 8 – Jul 10, 2026, Taipei, Taiwan, ROC.

C13. [MIDL'26] Skylar Stolte*†, Junfu Cheng*†, Chintan Acharya*, Lin Gu, Andrew O'Shea, Aprinda Indahlastari, Adam J. Woods, **Ruogu Fang**: BETA: Resting-state fMRI Biotypes for tDCS Efficacy in Anxiety among Older Adults at Risk for Alzheimer's Disease, Medical Imaging with Deep Learning (MIDL), Jul 8 – Jul 10, 2026, Taipei, Taiwan, ROC. (†: co-first authors)

C14. [SPIE'26] Zhimin Zong*, **Ruogu Fang**: Integrating Retinal Fundus Imaging and Computable Phenotypes for Subphenotype Discovery and LLM-Assisted Early and Late Onset Alzheimer's Diagnosis, SPIE Medical Imaging, Vancouver, Canada. Feb. 15-19, 2026.

C15. [AD'25] Tyler Ann Busch, Kevin Iversen, Skylar Stolte*, Gavin Hart*, Jason Chen*, Pratyush Shukla*, Samantha Pedersen, Andrew O'Shea, Yunfeng Dai, Alejandro Albizu, Adam J. Woods, **Ruogu Fang**, Aprinda Indahlastari: *Neuroanatomical Predictors of tDCS Current Distribution in Cognitively Healthy Older Adults*. Alzheimer's & Dementia, Vol. 21, e103172, December 2025. DOI: 10.1002/alz70856_103172.

C16. [ICCV'25] Shanlin Sun, Yifan Wang, Hanwen Zhang, Yifeng Xong, Qin Ren, **Ruogu Fang**†, Xiaohui Xie, Chenyu You†: Ouroboros: Single-step Diffusion Models for Cycle-consistent Forward and Inverse Rendering, International Conference on Computer Vision (ICCV), Oct 19 - Oct 25, 2025, Hawaii, USA. (Acceptance Rate=24%) (8 pages) (†: co-corresponding authors)

C17. [MICCAI'25] Qin Ren, Yifan Wang, **Ruogu Fang**, Haibin Ling, Chenyu You: OTSurv: A Novel Multiple Instance Learning Framework for Survival Prediction with Heterogeneity-aware Optimal Transport, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), September 23-28, Daejeon, Republic of Korea, 2025. (Acceptance Rate=29%) (10 pages)

C18. [ASTRO'25] Zhuobiao Qiao*, Weiguang Yao, Zeyu Zhang, Baoshe Zhang, Chris Beltran, Byong Yong Yi, Eric Paulson, Lei Ren, **Ruogu Fang**, Mi Huang: Improving Proton Dose Calculation Accuracy Using ROI-Focused AI Deep Learning Network for CBCT Based Adaptive Proton Therapy, American Society for Radiation Oncology (ASTRO), 2025.

C19. [BMES'25] Junfu Cheng, Aprinda Indahlastari, **Ruogu Fang**: Attention-Based Multimodal Modeling of Electric Field and Clinico-Demographic Features Predicts State Anxiety Response to tDCS in Older Adults. Biomedical Engineering Society (BMES) Annual Meeting, San Diego, October 2025.

C20. [BMES'25] Wasif Khan, **Ruogu Fang**: Physical Activity Supports Subcortical Brain Health in Older Adults: Insights from the UK Biobank. Biomedical Engineering Society (BMES) Annual Meeting, San Diego, October 2025.

C21. [BMES'25] Hart. G.M., Silva, G. L., Busch, T.A., Iversen, K. D., **Fang, R**., Woods, A., Queen, A. I. Impacts of Head Tissue Variability on Predicted Current Density in Aging Brains. Abstract accepted for poster presentation at the 2025 Biomedical Engineering Society (BMES) Annual Meeting, San Diego, CA. October 2025.

C22. [IBS'25] Skylar E. Stolte*, Aprinda Indahlastari, Alejandro Albizu, Adam J. Woods, **Ruogu Fang**: Building tDCS Digital Twins: AI-Powered Prediction of Personalized Electrical Field Maps from MRI, in 6th International Brain Stimulation Conference, Kobe, Japan. February 23–26, 2025. Brain Stimulation, volume 18, issue 1, pages 594-595 (2025). 10.1016/j.brs.2024.12.1101

C23. [IBS'25] Junfu Cheng*, Daniel Rodriguez*, Aprinda Indahlastari, Adam Woods, **Ruogu Fang**: Baseline Clinical Markers and Machine Learning-Based Prediction of tDCS Treatment Outcomes in Anxiety Disorders, in 6th International Brain Stimulation Conference, Kobe, Japan. February 23–26, 2025. Brain Stimulation, volume 18, issue 1, pages 595- (2025). 10.1016/j.brs.2024.12.1102

C24. [IBS'25] Aprinda Indahlastari, Alejandro Albizu, Skylar Stolte, Ruogu Fang, Adam Woods: Practical considerations in modeling tDCS in older adults, in 6th International Brain Stimulation Conference, Kobe, Japan. February 23–26, 2025. Brain Stimulation, volume 18, issue 1, pages 332-333 (2025). 10.1016/j.brs.2024.12.356

C25. [NCF SfN'25] Gavin Hart G, Giovana Silva, Tyler Busch, Kevin Iversen, Ruogu Fang, Adam Woods, Aprinda Indahlastari Queen: Impacts of Head Tissue Variability on Predicted Current Density in Aging Brains, North Central Florida SfN Conference, February 7th, 2025.

C26. [SPR'24] Gilbert, F., Yang, F., Farkas, A., Gardy, S., Engle, H., Cordova, A., Cheng, R., Sabatinelli, D., Ding, M., Fang, R., & Keil, A. Implementation of an AI Generated Affective Picture Set: Self-report and Psychophysiological Validation. Society for Psychophysiological Research Annual Convention. Prague, CZ. 2024.

C27. [BMES'24] Jason Chen*, Veronica Ramos*, Skylar Stolte*, Aprinda Indahlastari, Adam Woods, Ruogu Fang: Comparison of GRACE and SynthSeg Deep Learning Models for Whole-Head Brain Segmentation, in Biomedical Engineering Society Annual Meeting, Baltimore, MD. October 23-26, 2024.

C28. [BMES'24] Daniel Rodriguez*, Junfu Cheng*, Skylar Stolte*, Aprinda Queen, Adam Woods, **Ruogu Fang**: Baseline Clinical Characteristics and Machine Learning Predict tDCS Treatment Outcomes for Anxiety, in Biomedical Engineering Society Annual Meeting, Baltimore, MD. October 23-26, 2024.

C29. [VSS'24] Yujun Chen, Faith Gilbert, Ethan Smith, **Ruogu Fang**, Andreas Keil, Mingzhou Ding: Neural Responses to Natural Versus AI-generated Affective Images, Vision Sciences Society Annual Meeting, Journal of Vision, Vol. 24, 473, 2024. https://doi.org/10.1167/jov.24.10.473

C30. [NASEM'24] **Ruogu Fang**, et al., National Academies of Sciences, Engineering, and Medicine: Exploring the Bidirectional Relationship Between Artificial Intelligence and Neuroscience: Proceedings of a Workshop. Washington, DC: The National Academies Press, 2024. https://doi.org/10.17226/27764.

C31. [MICCAI'24] Skylar Stolte*, Aprinda Indahlastari, Alejandro Albizu, Adam Woods, **Ruogu Fang**. Towards tDCS Digital Twins using Deep Learning-based Direct Estimation of Personalized Electrical Field Maps from T1-Weighted MRI. Marrakesh, Morocco, October 6-10, 2024. (12 pages, early acceptance: 11%)

C32. [EMBS'24] Seowumg Leem*, Yunchao Yang, Adam J. Woods, **Ruogu Fang**: Deep Learning Analysis of Retinal Structures and Risk Factors of Alzheimer's Disease. 46th International Conference of the IEEE Engineering in Medicine and Biology Society, Orlando, FL. July 15-19, 2024. (IEEE EMBC NextGen Scholar Award, oral presentation)

C33. [HEALTHINF'24] Chaoyue Sun, Yiyang Liu, Christina Parisi, Rebecca Fisk-Hoffman, Marco Salemi, **Ruogu Fang**, Brandi Danforth, Mattia Prosperi and Simone Marini: Learning on Forecasting HIV Epidemic Based on Individuals'

Contact Networks. The 17th International Conference on Health Informatics (HEALTHINF/BIOSTEC), Rome - Italy, February 21 - 23, 2024, published in Proceedings of the 17th International Joint Conference on Biomedical Engineering Systems and Technologies, Volume 2, ISBN 978-989-758-688-0, ISSN 2184-4305, pages 103-111, 2024. DOI: https://doi.org/10.5220/0012375400003657 (**Best Paper Award, only 1 among all submission**)

C34. [BMES'23] Joseph Cox*, Yunchao Yang, Huiwen Ju, Justin Broce*, **Ruogu Fang**: Large Brain Models: Large-scale 3D Pretrained Deep Learning Models for Neuroimages, in Biomedical Engineering Society Annual Meeting, Seattle, WA. October 11-14, 2023.

C35. [BMES'23] Seowung Leem*, Andreas Keil, Peng Liu*, Mingzhou Ding, **Ruogu Fang**: Can Deep Convolutional Neural Network Associate Emotions with Gabor Patches? in Biomedical Engineering Society Annual Meeting, Seattle, WA. October 11-14, 2023.

C36. [BMES'23] Cameron Celeste, Dion Ming, Justin Broce, Diandra Ojo, Emma Drobina, Juan Gilbert, **Ruogu Fang**, Ivana Parker: Ethnic Disparity in Diagnosing Asymptomatic Bacterial Vaginosis Using Machine Learning, in Biomedical Engineering Society Annual Meeting, Seattle, WA. October 11-14, 2023.

C37. [BMES'23] Dion Ming, Cameron Celeste, Emma Drobina, Justin Broce, Diandra Ojo, **Ruogu Fang**, Ivana Parker: Feature selection in ML identifies unique bacteria to consider for fair BV diagnosis among ethnicities, in Biomedical Engineering Society Annual Meeting, Seattle, WA. October 11-14, 2023.

C38. [MICCAI'23] Skylar E. Stolte*, Kyle Volle, Aprinda Indahlastari, Alejandro Albizu, Adam J. Woods, Kevin Brink, Matthew Hale, **Ruogu Fang**: DOMINO++: Domain-aware Loss Regularization for Deep Learning Generalizability, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Vancouver, Canada, October 8-12, 2023. (acceptance rate=32%)

C39. [MDS'23] Joshua Wong, Sina Aghili-Mehrizi, Reza Forghani, Abbas Babajani, Michael Okun, Christopher Butson, **Ruogu Fang**. AI-guided microelectrode recording for GPi DBS in Parkinson's Disease, International Congress of Parkinson's Disease and Movement Disorders, 2023.

C40. [AAPM'23] Zhuobiao Qiao[G], Zeyu Zhang, Zhuoran Jiang, Youfang Lai, Jessica Lee, Dapeng Oliver Wu, Chris Beltran, **Ruogu Fang**, Lei Ren, Mi Huang. Attention Module Embedded Generative Adversarial Network for Enhancing 3D CBCT Image Quality for Radiomics Analysis, AAPM Annual Conference, Imaging Snap Oral Session, Houston, TX, July 2023.

C41. [SPR'23] Faith E Gilbert, Hannah M Engle, Caitlin M Traiser, Arash Mirifar, Christian Panitz, Jourdan Pouliot, Laura C Ahumada Hernandez, Ethan Smith[U], **Ruogu Fang**, Mingzhou Ding, Andreas Keil: Building and Evaluating an AI-Generated, Standardized Affective Picture Dataset, Annual meeting of the Society for Psychophysiological Research, New Orleans LA, September 27 to October 1, 2023.

C42. [ICDCS'23] Hong Huang[*], Lan Zhang, Chaoyue Sun[*], **Ruogu Fang**, Xiaoyong Yuan and Dapeng Wu: Distributed Pruning Towards Tiny Neural Networks in Federated Learning, the 43rd IEEE International Conference on Distributed Computing Systems (ICDCS 2023), July 18-21, 2023, Hong Kong (acceptance ratio: 18.9%)

C43. [RSNA'22] Garrett Fullerton*, Simon Kato*, Dhanashree Rajderkar, John Rees, Pina Sanelli, **Ruogu Fang**: MAGIC: Multitask synthesis of contrast-free CT perfusion maps via generative adversarial network, in Annual Meeting of Radiology Society of North America (RSNA), Chicago, Nov. 27-Dec. 1, 2022.

C44. [ASRA'22] Jin Choi, Angela Fadil, Caitlin M Gray, Terrie Vasilopoulos, Jiaqing Zhang[G], **Ruogu Fang**, Sanjeev Kumar. Analysis of Procedural Accuracy in Lumbar Transforaminal Epidural Steroid Injections, 21st ASRA Annual Pain Medicine Conference, Orlando, FL. November 17-19, 2022.

C45. [SfN'22] Seowung Leem*, Andreas Keil, Peng Liu*, Mingzhou Ding, **Ruogu Fang**: Differential Aversive and Appetitive Conditioning in Artificial Neural Networks, in Society of Neuroscience Annual Meeting, San Diego, CA. November 12, 2022.

C46. [Diabetes'22] Yehua Wang, Ping Zhang, **Ruogu Fang**, Joshua Brown, Jingchuan Guo, Tianze Jiao, Hui Shao: 173-OR: Associations between Post-Discharge Care (PDC) and Cognitive Impairment-Related Hospital Readmissions for Ketoacidosis and Severe Hypoglycemia in Adults with Diabetes, *Diabetes* 71, no. Supplement_1, 2022.

C47. [Diabetes'22] Dawei Guan, Ping Zhang, Shichao Tang, Joshua Brown, **Ruogu Fang**, Jingchuan Guo, Yongkang Zhang, Hui Shao: 893-P: Incremental Health Care Utilizations and Medical Expenditures Associated with Cognitive Impairment among Older Adults with Diabetes, in *Diabetes* 71, no. Supplement_1, 2022.

C48. [MICCAI'22] Skylar Stolte*, Kyle Volle, Aprinda Indahlastari, Alejandro Albizu, Adam Woods, Kevin Brink, Matthew Hale, **Ruogu Fang**: DOMINO: Domain-aware Model Calibration in Medical Image Segmentation, 25th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Singapore, Sep. 18-22, 2022. (**Best Paper Presentation Award Runner Up, Women in MICCAI.** Oral Presentation rate=2.3%, Early acceptance rate=13%) (12 pages)

C49. [MAIN'21] Peng Liu*, Ke Bo, Lihan Cui, Yujun Chen, Charlie Tran*, Mingzhou Ding†, **Ruogu Fang**†: A deep neural network model for emotion perception, in *Montreal AI & Neuroscience (MAIN),* November 29-30, 2021. († co-corresponding authors) **Best Abstract Award** (undergraduate and graduate level)

C50. [SfN'21] Peng Liu*, Ke Bo, Lihan Cui, Yujun Chen, Charlie Tran*, Mingzhou Ding†, **Ruogu Fang**†: Emergence of emotion selectivity in deep neural networks trained to recognize visual objects, in *Annual Meeting of Society of Neuroscience*, November 8-16, 2021. († co-corresponding authors)

C51. [SfN'21] Peng Liu*, Ke Bo, Lihan Cui, Yujun Chen, Charlie Tran*, **Ruogu Fang**†, Mingzhou Ding†: A deep neural network model for emotion perception, in *Annual Meeting of Society of Neuroscience,* November 8-16, 2021. († co-corresponding authors)

C52. [SfN'21] Alejandro Albizu, **Ruogu Fang**, Aprinda Indahlastari, Andrew O'Shea, Skylar E. Stolte*, Kyle B. See*, Emanuel M. Boutzoukas, Jessica N. Kraft, Nicole R. Nissim, and Adam J. Woods: Machine-Learning Defined Precision tDCS for Improving Cognitive Function, in *Annual Meeting of Society of Neuroscience*, November 8-16, 2021.

C53. [INS'21] Alejandro Albizu, **Ruogu Fang**, Aprinda Indahlastari, Andrew O'Shea, Skylar E. Stolte*, Kyle B. See*, Emanuel M. Boutzoukas, Jessica N. Kraft, Nicole R. Nissim, and Adam J. Woods: Individualized Machine-Learning Derived Transcranial Electrical Stimulation Optimization for Working Memory Improvement in Older Adults. 49th Annual Meeting of the International Neuropsychological Society.

C54. [BMES'21] Garrett Fullerton*, Simon Kato*, **Ruogu Fang**: MAGIC: Multitask Automated Generation of Inter-modal CT Perfusion Maps via Generative Adversarial Network, in *Biomedical Engineering Society Annual Meeting*, Orlando, FL. October 6-9, 2021.

C55. [BMES'21] Kyle See*, Rachel Judy, Stephen Coombes, **Ruogu Fang**: Predicting Treatment Outcome in Spinal Cord Stimulation with EEG, in *Biomedical Engineering Society Annual Meeting*, Orlando, FL. October 6-9, 2021.

C56. [BMES'21] Hely Lin*, **Ruogu Fang**: Ensemble Machine Learning for Alzheimer's disease Classification from Retinal Vasculature, in *Biomedical Engineering Society Annual Meeting*, Orlando, FL. October 6-9, 2021.

C57. [HealthNLP'21] Zehao Yu, Xi Yang, Gianna L Sweeting*, Yinghan Ma, Skylar E Stolte*, **Ruogu Fang**, Yonghui Wu: Identify Diabetic Retinopathy-related Clinical Concepts Using Transformer-based Natural Language Processing Methods, in *IEEE 9th International Conference on Healthcare Informatics (ICHI)*, Victoria, Canada (pp. 499-500), August 9, 2021.

C58. [RSNA'20] Maximillian Diaz*, Jianqiao Tian*, **Ruogu Fang**: Machine Learning for Parkinson's Disease Diagnosis Using Fundus Eye Images, *Annual Meeting of Radiology Society of North America (RSNA),* December 2020. (Oral Presentation) **Featured by Forbes, RSNA, Diagnostic Imaging, Medscape Medical News, American Journal of Managed Care, SciTech Daily, Applied Radiology The Journal of Practical Medical Imaging and Management, and 50+ media outlets.**

C59. [APS'20] Bhargav Sriram Siddani, S. Balachandar, **Ruogu Fang**, William Chandler Moore, Yunchao Yang: Dispersed Multiphase Flow Generation using 3D Steerable Convolutional Neural Network, in *73rd Annual Meeting of the American Physical Society Division of Fluid Dynamics,* Volume 65, Number 13, Virtual, CT (Chicago Time), November 22-24, 2020.

C60. [AHA'20] Skylar Stolte*, Yonghui Wu, William R Hogan, Yan Gong, **Ruogu Fang**: Artificial Intelligence For Characterizing Heart Failure In Cardiac Magnetic Resonance Images, in *American Heart Association Scientific Sessions*, November 13-17, 2020.

C61. [NNC'20] Albizu A, **Fang R**, Indahlastari A, Nissim NR, O Shea A, Woods AJ: Determinants of Treatment Response to Transcranial Direct Current Stimulation, in *the 5th Annual NYC Neuromodulation Conference*, April 2020. **Outstanding Presentation by Early Career Scientist Award**

C62. [INS'20] Alejandro Albizu, **Ruogu Fang**, Aprinda Indahlastari, Nicole R. Nissim, Andrew OShea, Adam J. Woods: Building Personalized Medicine Models for Therapeutic Applications of Transcranial Electrical Stimulation, in *the 48th Annual Meeting of the International Neuropsychological Society* in February 2020.

C63. [SIIM'20] Yao Xiao*, Manuel M. Arreola, Izabella Barreto, Wesley E. Bolch, W. Christopher Fox, Keith Peters, Dhanashree A. Rajderkar, John H. Rees, **Ruogu Fang**: Multi-Series CT Image Super-Resolution by using Transfer Generative Adversarial Network, in *Society for Imaging Informatics in Medicine (SIIM) Annual Meeting*, Austin, Texas, June 24-26, 2020 (Oral)

C64. [ISBI'20] Yao Xiao*, Keith R. Peters, W. Christopher Fox, John H. Rees, Dhanashree A. Rajderkar, Manuel M. Arreola, Izabella Barreto, Wesley E. Bolch, **Ruogu Fang**: Transfer-GAN: Multimodal CT Image Super-Resolution via Transfer Generative Adversarial Networks, in *IEEE International Symposium on Biomedical Imaging (ISBI)* (pp. 195-198), Iowa City, Iowa, April 3-7, 2020.

- Travel Awards funded by the US National Institutes of Health (NIH), National Institute of Biomedical Imaging and Bioengineering (NIBIB), National Cancer Institute (NCI), and Graduate Student Council)

C65. [SPIE Medical Imaging'20] Yao Xiao* and **Ruogu Fang**: Transfer Generative Adversarial Network for Multimodal CT Image Super-Resolution, in *SPIE Medical Imaging* (vol. 11313, p.1131306), Houston, Texas, Feb 15-20, 2020 (Oral)

C66. [SNAMC'20] Justin L Brown, Daniel El Basha*, Nathalie Correa, Yao Xiao*, Izabella Barreto, **Ruogu Fang**, Chan Kim, Wesley E. Bolch: Monte Carlo Dosimetry For CT Brain Perfusion Studies Utilizing Volumetric Acquisitions, in *Joint International Conference on Supercomputing in Nuclear Applications + Monte Carlo*, 2020.

C67. [ASCPT'20] Marwa Tantawy, Sonal Singh, Guang Yang, Matt Gitzendanner, Yiqing Chen, Yonghui Wu, **Ruogu Fang**, William Hogan, Yan Gong: ZMAT4 and DOCK9 Variants Associated with Heart Failure in Breast Cancer Patients in the UK Biobank data, in *American Society for Clinical Pharmacology and Therapeutics Annual Meeting*, Houston, TX, March 18-21, 2020. **David Goldstein M.D., Ph.D., Presidential Trainee Award for the highest-scoring abstract 2020** (This award is presented each year to recognize the highest scoring trainee abstract.)

C68. [RSNA'19] Yao Xiao*, Manual Arreola, Izabella Barreto, W. Christopher Fox, Keith Peters, **Ruogu Fang**: Multimodal CT Image Super-Resolution via Transfer Generative Adversarial Network, in *Annual Meeting of Radiology Society of North American*, December 2019. (Oral presentation)

C69. [MICCAI'19] Peng Liu*, Bin Kong, Zhongyu Li, Shaoting Zhang, **Ruogu Fang**: CFEA: Collaborative Feature Ensembling Adaptation for Domain Adaptation in Unsupervised Optic Disc and Cup Segmentation, in the *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pp. 521-529, October 2019. (Early Acceptance Rate = 10%-15%)

C70. [MICCAI'19] Siyuan Pan, Yuxin Xue, Bin Sheng, Xuhong Hou, Huating Li, **Ruogu Fang**, Weiping Jia, Jing Qin: Abdominal Adipose Tissue Segmentation in MRI with Double Loss Function Collaborative Learning, in *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pp. 41-49, October 2019. (Early Acceptance Rate = 10%-15%)

C71. [BMES'19] Yao Xiao*, **Ruogu Fang**: Multimodal CT Image Super-Resolution via Transfer-GAN, in *Biomedical Engineering Society Annual Meeting*, October 2019, Philadelphia, PA.

C72. [BMES'19] Jianqiao Tian*, Max Diaz*, **Ruogu Fang**: Deep Learning-based Alzheimer's Disease Classification of FDG-PET and AV45 PET Images, in *Biomedical Engineering Society Annual Meeting*, October 2019, Philadelphia, PA.

C73. [BMES'19] Skylar Stolte*, **Ruogu Fang**: Artificial Intelligence For Automated Diagnosis of Glaucoma In Stereoscopic Images, in *Biomedical Engineering Society Annual Meeting*, October 2019, Philadelphia, PA.

C74. [BMES'19] Kyle See*, **Ruogu Fang**: Classification Of Neural Stimulations In The Brain With Super Voxels, in *Biomedical Engineering Society Annual Meeting*, October 2019, Philadelphia, PA.

C75. [BMES'19] Skylar Stolte*, Kyle See*, Daniel El Basha*, **Ruogu Fang**: Retinal Disease Diagnosis Using Mobile Devices, in *Biomedical Engineering Society Annual Meeting*, October 2019, Philadelphia, PA.

C76. [MICCAI'18] Peng Liu*, Yangjunyi Li, Mohammad D El Basha, **Ruogu Fang**: Neural Network Evolution Using Expedited Genetic Algorithm for Medical Image Denoising, in *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pp. 12-20, September 2018, Granada, Spain.

C77. [MICCAI'18] Yaxin Shen, **Ruogu Fang**, Bin Sheng, Ling Dai, Huating Li, Jing Qin, Qiang Wu, Weiping Jia: Multi-task Fundus Image Quality Assessment via Transfer Learning and Landmarks Detection, pp.28-36, *Machine Learning in Medical Imaging*, September 2018, Granada, Spain.

C78. [BMES'18] Yao Xiao*, Peng Liu, **Ruogu Fang**: Low-Dose CT Perfusion Image Restoration and Radiation Reduction, in *Biomedical Engineering Society Annual Meeting*, Atlanta, GA, October 2018. (Oral Presentation)

C79. [BMES'18] Yao Xiao*, Yun Liang, Yunmei Chen, Xiaojing Ye, and **Ruogu Fang**: Efficient Multi-Modality Medical Image Joint Reconstruction via Vectorized Gradient, in *Biomedical Engineering Society Annual Meeting*, Atlanta, GA, October 2018.

C80. [BMES'18] Skylar Stolte*, Yao Xiao*, **Ruogu Fang**: Multi-Modality Brain Image Co-Registration, in *Biomedical Engineering Society Annual* Meeting, Atlanta, GA, October 2018.

C81. [BMES'18] Kyle B. See*, **Ruogu Fang**: Decision Tree-based Classification for Differentiating System Lupus Erythematosus and Mixed Connective Tissue Disease, in *Biomedical Engineering Society Annual Meeting*, Atlanta, GA, October 2018.

C82. [TAPIA'18] Yao Xiao*, Peng Liu*, Yun Liang*, **Ruogu Fang**: STDN: Spatial-Temporal De- noising Net for Radiation Optimization in CT Perfusion, in *ACM Richard Tapia Celebration of Diversity in Computing*, September, Orlando, FL. 2018. **Student Travel Scholarship**

C83. [ASNR'18] Yao Xiao*, Yun Liang*, Yunmei Chen, Xiaojing Ye, **Ruogu Fang**: Multi-Modality PET-MRI Image Joint Reconstruction, in *ASNR 56th Annual Meeting & The Foundation of the ASNR Symposium*, Vancouver, Canada, June 2018. (Oral Presentation)

C84. [ASNR'18] Yao Xiao*, Pina C. Sanelli, **Ruogu Fang**: Image Super-Resolution and Radiation Reduction via Deep Learning, in *ASNR 56th Annual Meeting & The Foundation of the ASNR Symposium*, Vancouver, Canada, June 2018. (Oral Presentation)

C85. [ASNR'18] Peng Liu*, **Ruogu Fang**: Regulated-convolutional Networks for Low-dose Cerebral CT Perfusion Restoration, in *ASNR 56th Annual Meeting & The Foundation of the ASNR Symposium*, Vancouver, Canada, June 2018. (Oral Presentation)

C86. [ISBI'18] Peng Liu*, **Ruogu Fang**: SDCNET: Smoothed Dense-Convolution Network For Restoring Low-Dose Cerebral CT Perfusion, in *IEEE International Symposium on Biomedical Imaging*, pp. 349-352, April 2018, Washington D.C.

C87. [MICCAI'17] Yao Xiao*, **Ruogu Fang**: STAR: Spatio-Temporal Architecture for super-Resolution in Low-Dose CT Perfusion, in *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Machine Learning in Medical Imaging Workshop*, pp. 97-105, September 2017, Quebec, Canada. **MICCAI Student Travel Award, 50/810=6% award rate**

C88. [BMES'17] Yao Xiao*, **Ruogu Fang**: Cardiovascular Disease Prediction and Risk Factor Mining with RFMiner, in *Biomedical Engineering Society Annual Meeting (BMES) Annual Meeting*, October 11-14, 2017, Phoenix, Arizona.

C89. [BMES'17] Yao Xiao*, **Ruogu Fang**: Accelerated Brain Perfusion Imaging via Spatio-Temporal Super-Resolution, in *Biomedical Engineering Society Annual Meeting (BMES) Annual Meeting,* October 11-14, 2017, Phoenix, Arizona. (oral presentation)

C90. [BMES'17] Peng Liu*, **Ruogu Fang**: A Simple and Realistic Simulation Method for Low-Dose CT., in *Biomedical Engineering Society Annual Meeting (BMES) Annual Meeting*, October 11-14, 2017, Phoenix, Arizona.

C91. [CHASE'17]) Yao Xiao*, **Ruogu Fang**: RFMiner: Risk Factors Discovery and Mining for Preventive Cardiovascular Health, in *the Second IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies*, pp. 278-279, July 2017, Philadelphia, USA. (**NSF Student Travel Award**)

C92. [NIPS WiML'16] Maryamossadat Aghili*, **Ruogu Fang**: Towards High-Throughput Abnormal Brain Screening in MRI Images, in *Women in Machine Learning Workshop, Neural Information Processing Systems (NIPS)*, December 2016, Barcelona, Spain.

C93. [BMES'16b] Paul Naghshineh*, Peng Liu*, **Ruogu Fang**: CT Perfusion Image Super-Resolution Using a Deep Convolutional Network, in *Biomedical Engineering Society Annual Meeting (BMES),* October 5-8, 2016 in Minneapolis, Minnesota. (NSF-REU Best Post Award, SCIS FIU)

C94. [BMES'16a] Anuradha Godavarty, Rebecca Kwasinki, Cristianne Fernandez, Yuanyuan Zhu*, Edwin Robledo, F. Perez-Clavijo, **Ruogu Fang**: Physiological Assessment of Wound Healing using a Near-Infrared Optical Scanner, in *Biomedical Engineering Society Annual Meeting (BMES),* October 5-8, 2016 in Minneapolis, Minnesota.

C95. [ISBI'16c] **Ruogu Fang,** Xing Pang*, Arash Dadkhah, Jiali Lei, Elizabeth Solis, Suset Rodriguesz, Francisco Perez-Clavijo, Stephen Wigley, Charles Buscemi, Anuradha Godvarty: Automatic Segmentation of Lower Extremity Ulcers in Near-Infrared Optical Imaging, in *IEEE International Symposium on Biomedical Imaging (ISBI)*, Prague, Czech Republic, April 2016.

C96. [ISBI'16b] Zhongyu Li, Fumin Shen, **Ruogu Fang**, Sailesh Conjeti, Amin Katouzian, Shaoting Zhang: Maximum Inner Product Search for Morphological Retrieval of Large-Scale Neuron Data, in *IEEE International Symposium on Biomedical Imaging (ISBI)*, pp. 602-606, Prague, Czech Republic, April 2016. (Oral Presentation Rate: 19%)

C97. [ISBI'16a] **Ruogu Fang**, Ajay Gupta, Pina C. Sanelli: Direct Estimation of Permeability Maps for Low-Dose CT Perfusion, in *IEEE International Symposium on Biomedical Imaging (ISBI)*, pp. 739-742, Prague, Czech Republic, April 2016.

C98. [OSA'16] Xing Pang*, Arash Dadkhah, Jaili Lei, Elizabeth Solis, Suset Rodriguez, Francisco Perez- Clavijo, Stephen Wigley, **Ruogu Fang**, Anuradha Godvarty: Near-Infrared Optical Imaging and Wound Segmentation in Lower Extremity Ulcers, in *Optical Society of America Annual Meeting (OSA),* pp. JTu3A-43, 2016.

C99. [SPIE'16] Arash Dadkhah, Xing Pang*, Elizabeth Solis, **Ruogu Fang**, Anuradha Godvarty: Wound Size Measurement of Lower Extremity Ulcers Using Segmentation Algorithms, in *SPIE Proceedings in Photonics West*, vol. 9703, p. 97031D, San Francisco, February 2016.

C100. [WI'16] **Ruogu Fang**, Xing Pang*, Arash Dadkhah, Jiali Lei, Elizabeth Solis, Suset ROdriguez, Francisco Perez-Calvijo, Stephen Wigley, Charles Buscemi, Anuradha Godvarty, in *Wound Segmentation in Near-Infrared Optical Imaging, in Innovation in Wound Healing*, Hawks Cay, FL. 2015.

C101. [WI'16] Rebecca Kwasinski, Cristianne Fernandez, Kevin Leiva, Edwin Robledo, Yuanyuan Zhu, Penelope Kallis, Francesco-Perez Clavijo, **Ruogu Fang**, Robert Kirsner, Anuradha Godavarty: Hemodynamic Imaging of Lower Extremity Ulcers, *Innovations in Wound Healing*, 2016.

C102. [MICCAI'15] Ruoyu Li, Yeqing Li, **Ruogu Fang**, Shaoting Zhang, Junzhou Huang: Fast Preconditioning for Accelerated Multi-Contrast MRI Reconstruction, in *the 18th Annual International Conference on Medical Image*

Computing and Computer Assisted Intervention (MICCAI), pp. 700-707, Munich, Germany, Oct. 5-9, 2015. (Oral Presentation Rate: 4%)

C103. [MICCAI-MCV'15] **Ruogu Fang**, Ming Ni, Junzhou Huang, Qianmu Li, Tao Li: Efficient 4D Non-Local Tensor Total-Variation for Low-Dose CT Perfusion Deconvolution, *The 18th Annual International Conference on Medical Image Computing and Computer Assisted Intervention, Workshop on Medical Computer Vision*, pp. 168-179, Munich, Germany, October 2015.

C104. [BMES'15] **Ruogu Fang**, Ming Ni, Junzhou Huang, Qianmu Li, Tao Li: Robust Low-Dose CT Perfusion Deconvolution via Non-Local Tensor Total Variation, in *Biomedical Engineering Society Annual Meeting (BMES),* Tampa, FL, October 2015.

C105. [ISMRM'15] **Ruogu Fang**: 4-D Spatio-Temporal MR Perfusion Deconvolution via Tensor Total Variation, in *International Society for Magnetic Resonance in Medicine Annual Meeting* 2015 (Oral presentation, corresponding author).

C106. [ISBI'15b] Menglin Jiang, Shaoting Zhang, **Ruogu Fang**, Dimitris Metaxas: Leveraging Inverted Multi-Index for Scalable Retrieval of Mammographic Masses, in *IEEE International Symposium On Biomedical Imaging: From Nano To Macro*, pp. 276-280, New York, NY April 2015. Oral Presentation Rate: 18%

C107. [ISBI'15a] **Ruogu Fang**, Junzhou Huang, Wen-Ming Luh: A Spatio-Temporal Low-Rank Total Variation Approach For Denoising Arterial Spin Labeling MRI Data, in *IEEE International Symposium On Biomedical Imaging: From Nano To Macro*, pp. 154-161, New York, NY April 2015.

C108. [MICCAI'14] **Ruogu Fang**, Pina Sanelli, Shaoting Zhang, Tsuhan Chen: Tensor Total-Variation Regularized Deconvolution for Efficient Low-Dose CT Perfusion, in *The 17th Annual International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pp. 89-96, Boston, MA. September 2014. **MICCAI Student Travel Award**

C109. [MICCAI-STMI'14] **Ruogu Fang**, Tsuhan Chen, Pina C. Sanelli: Anisotropic Tensor Total Variation Regularization For Low Dose Low CT Perfusion Deconvolution, in *The 17th Annual International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Workshop on Sparsity Techniques in Medical Imaging*, Boston, MA. September 2014.

C110. [MICCAI'13] **Ruogu Fang**, Tsuhan Chen, Pina Sanelli: Tissue-Specific Sparse Deconvolution for Low-Dose CT Perfusion, in *The 16th Annual International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Japan, 2013.

C111. [ICIP'13] **Ruogu Fang**, Andrew C. Gallagher, Tsuhan Chen, Alexander Loui: Kinship Classification by Modeling Facial Feature Heredity, in *IEEE International Conference on Image Processing*, pp. 2983-2987, Melbourne, Australia, 2013 (Oral presentation)

C112. [MICCAI'12] **Ruogu Fang**, Tsuhan Chen, Pina Sanelli: Sparsity-Based Deconvolution of Low-Dose Perfusion CT Using Learned Dictionaries, in *The 15th Annual International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Nice, France, Lecture Notes in Computer Science Volume 7510, pp. 272-280, 2012. (corresponding author)

C113. [ISBI'12] **Ruogu Fang**, Tsuhan Chen, Pina Sanelli: Sparsity-Based Deconvolution Of Low-Dose Brain Perfusion CT In Subarachnoid Hemorrhage Patients, in *The 9th IEEE International Symposium on Biomedical Imaging*, pp. 872-875, 2012 **Oral Presentation Rate: 18%**

C114. [SPIE'12] **Ruogu Fang**, Ashish Raj, Tsuhan Chen, Pina C. Sanelli: Radiation dose reduction in computed tomography perfusion using spatial-temporal Bayesian methods, in *Proceedings of SPIE Medical Imaging*, Volume 8313, Paper 831345, 2012. (corresponding author)

C115. [MICCAI-AI'11] **Ruogu Fang**, Ramin Zabih, Ashish Raj, Tsuhan Chen: Segmentation of Liver Tumor Using Efficient Global Optimal Tree Metrics Graph Cuts, in Abdominal Imaging, in the *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI),* pp. 51-59, 2011 (Oral presentation, corresponding author)

C116. [ICIP'10] **Ruogu Fang**, Kevin D. Tang, Noah Snavely, Tsuhan Chen: Towards Computational Models of Kinship Verification, in *The 17th IEEE International Conference on Image Processing*, pp. 1577-1580, Hong Kong, 2010. Oral presentation, **Best Paper Award, 1/1190 accepted papers**

C117. [WNYIPW'10] **Ruogu Fang**, Joyce Yu-hsin Chen, Ramin Zabih, Tsuhan Chen: Tree-Metrics Graph Cuts For Brain MRI Segmentation With Tree Cutting, in *IEEE Western New York Image Processing Workshop*, pp. 10-13, 2010 (Oral presentation).

C118. [VISAPP'09] Chongyang Liu, **Ruogu Fang**, Nelson H.C.Yung: Adaptive Scale Robust Feature Density Approximation For Visual Object Representation And Tracking, pp. 535-540, in *IEEE International Conference on Computer Vision Theory and Applications*, Lisboa, Portugal, 2009.

INVITED TALKS, PRESENTATIONS, AND WEBINARS

Summary: 56 invited talks in total, including NAM, Harvard, Stanford, UCB, UCSF, UCLA, U Tokyo, RIKEN, UBC, Oxford, U Marrakesh, NASEM, and GRC

International and National Talks

T1. **Montreal Artificial Intelligence and Neuroscience (MAIN) Conference (Invited Talk),** "Bridging Minds and Machines: From Neural Representations to Brain-Inspired Learning", virtual, Oct. 29, 2026

T2. **Annual Meeting of Biomedical Engineering Society (BMES) (Invited Talk),** "From Data to Decisions: Brain Digital Twins, Foundation Models and Agentic AI", Orlando, FL. Oct. 22, 2026

T3. **Vanderbilt's third annual AI Summer School**, "Two-Way Intelligence: What Brains Teach AI and AI Teaches Us About the Brain", August 13, 2026

T4. **National Academy of Medicine**, Victor J. Dzau Emerging Leaders in Health and Medicine Meeting, "When Brain Meets AI", July 14, 2026

T5. **Harvard University, Center for Machine Learning Seminar Series,** "A Tale of Two Frontiers: Intelligence in Nature, Insight in Machines," virtual, Dec. 2025

T6. **Machine Learning Medical Imaging Consortium (MaLMIC),** University of Toronto and Ontario Institute for Cancer Research, "Brain Digital Twins and Foundation Models". Canada, virtual, April, 2026

T7. **Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital (MGH) & HMS & MIT, "**New Vision into Brain Health through Multimodal Digital Twins and Medical Foundation Models", Boston, July 2025.

T8. **Harvard Medical School and Massachusetts General Brighton (MGB)**, Department of Radiology, "When Brain Meets AI - Natural Intelligence-Inspired AI and AI-Empowered Brain Health", Boston, June 2025.

T9. **Stanford University**, Stat4Onc Short Course, **"**Unlocking Brain Health through Multimodal Digital Twins and 3D Vision Foundation Models", Virtual, May 2025 (CME credit)

T10. **Stanford University**, Stanford Science Fellow Program, Palo Alto, CA. "Strategic Career Development for the Modern Scientists", February 2025

T11. **Stanford University**, Department of Radiology, Palo Alto, CA. "A Tale of the Brain and AI: Multimodal Digital Twins and 3D Vision Foundation Models", February 2025

T12. **Stanford University**, Stanford Science Fellow Program, Palo Alto, CA. "Peek into the Brain through AI: Multimodal Digital Twins and 3D Vision Foundation Models", February 2025

T13. **Stanford University**, Department of Radiation Oncology, Palo Alto, CA. "Unlocking Brain Health through Multimodal Digital Twins and 3D Vision Foundation Models", January 2025

T14. **UC Berkely-UC San Francisco Computational Precision Health (CPH) Seminar (virtual), "**Unlocking Brain Health through Multimodal Digital Twins and 3D Vision Foundation Models", January 2025

T15. **University of California at Los Angeles (UCLA) BIOE Departmental Seminar Series, "**A Tale of Two Frontiers: When Brain Meets AI: Multimodal Digital Twins and Foundation Models", November 2024

T16. **Special Session on Multimodal AI in Biomedical Data Analysis and Disease Diagnosis, "**Multimodal Digital Twins to Unlock Brain Health", Annual Meeting of Biomedical Engineering Society, Baltimore, MD, October 2024

T17. **University of Marrakesh,** Marrakesh, Morocco, "NeuroAI: Brain-Inspired AI and AI-Empowered Precision Brain Health", October 2024

T18. **Human-Centered AI Conference**, "Brain-Inspired Affective AI – Advancing Human-Centered AI Through Emotion Processing and Associative Learning", Malibu, CA, September 2024.

T19. **University of Tokyo and RIKEN,** Tokyo, Japan. "Natural Intelligence-Inspired AI and AI-Empowered Brain Health", May 2024

T20. **AI4Health Conference,** Orlando, FL. "AI4Brain: New Frontiers of Neuroscience and Brain Health", April 2024 (CME)

T21. **National Academy (NASEM),** Session on the Impact of AI in Clinical and Healthcare Environment, Washington, DC, "Exploring the Bidirectional Relationship between Artificial Intelligence and Neuroscience", March 2024

T22. **Academy of Science, Engineering & Medicine of Florida (ASEMFL**), Orlando, FL. "A Tale of Two Frontiers: When Brain Meets AI – Natural Intelligence-Inspired AI and AI-Empowered Brain Health (Rising Star Talk)", November 3, 2023

T23. **3rd Annual AI Symposium at UF Health Cancer Center: Insights in Cancer Imaging, "**AI for Medical Image Analysis 101", October 24, 2023, Gainesville, FL

T24. **University of British Columbia**, Pre-MICCAI Workshop, Vancouver, CA, "A Tale of Two Frontiers – When Brain Meets AI", October 7, 2023,

T25. **MRI Research Institute (MRIRI), Weill Cornell Medical College, Cornell University, NY, "**Generative, Trustworthy, and Precision AI in Radiology", August 7, 2023

T26. **Nvidia Artificial Intelligence Technology Center (NVAITC),** Trustworthy AI and Large Vision Models for Neuroimages", May 2023

T27. **Intelligent Critical Care Center Seminar Series**, Virtual, "Artificial Intelligence for Cognitive Aging: Novel Diagnosis and Personalized Intervention", March 16, 2023

T28. **Neural Information Processing System (NeurIPS), New Orleans, LA,** Medical Imaging meets NeurIPS workshop (Keynote Talk), **"**A Tale of Two Frontiers – When Brain Meets AI", December 2, 2022

T29. **Stanford University, College of Medicine**, Virtual, "Artificial Intelligence in Cognitive Aging and Brain-Inspired AI", November 11, 2022

T30. **Gordon Research Conference – Image Science**, Newry, Maine, USA, Emerging Imaging Techniques at the Intersection of Physics and Data Science, "From Zero to One: Physiology-Informed Deep Learning for Contrast-Free CT Perfusion in Stroke Care", June 2022

T31. **Gordon Research Conference – System Aging**, Newry, Maine, USA, Systemic Processes, Omics Approaches and Biomarkers in Aging, "Artificial Intelligence and Machine Learning for Cognitive Aging: Novel Diagnosis and Precision Intervention", June 2022

T32. **University of Florida Center for Cognitive Aging and Memory Research Day**, Gainesville, FL. "Artificial Intelligence for Novel Diagnosis of Alzheimer's Disease from Retinal Imaging", May 2022

T33. **University of Florida Biomedical Engineering Department Seminar Series**, Virtual, "Artificial Intelligence in Cognitive Aging – Novel Diagnosis and Precision Intervention", April 2022

T34. **University of Oxford "Artificial Intelligence for Mental Health" seminar series,** Virtual, "Modular machine learning for Alzheimer's disease classification from retinal vasculatures", January 2022

T35. **University of Florida Institute of Aging Seminar Series,** Virtual, "Artificial Intelligence in Cognitive Aging – Novel Diagnosis and Precision Intervention", January 2022

T36. **National Science Foundation- Cleveland Clinic Workshop on the Present and the Future of Artificial Intelligence in Biomedical Research**, Virtual, "AI in Precision Cognitive Aging Diagnosis and Intervention", June 2021

T37. **Annual Meeting of Radiology Society of North America (RSNA)**, Chicago, IL, "Multimodal CT Image Super-Resolution via Transfer Generative Adversarial Network", December 2019

T38. **Annual Meeting of the American Society of Neuroradiology (ASNR)**, Vancouver, Canada, "Multi-Modality PET-MRI Image Joint Reconstruction", June 2018

T39. **Annual Meeting of the American Society of Neuroradiology (ASNR)**, Vancouver, Canada, "Image Super-Resolution and Radiation Reduction via Deep Learning", June 2018

T40. **Annual Meeting of the American Society of Neuroradiology (ASNR)**, Vancouver, Canada, "Regulated-convolutional Networks for Low-dose Cerebral CT Perfusion Restoration", June 2018

T41. **ISMRM**, Toronto, Canada, "4-D Spatio-Temporal MR Perfusion Deconvolution via Tensor Total Variation", June 2015

T42. **Annual Meeting of Biomedical Engineering,** Tampa, FL. "Robust Low-Dose CT Perfusion Deconvolution via Non-Local Tensor Total Variation", October 2015

T43. **International Conference on Computational Advances in Bio and Medical Sciences,** Robust Medical Image Analysis In Assessing Disease Progression and Treatment Response", June 2014

Campus and Departmental Talks

T44. **Vanderbilt University Institute of Imaging Science (VUIIS) Retreat (Invited Talk),** "From Images to Intelligence: Foundation Models, Digital Twins, and Agentic AI", Vanderbilt Health, Oct. 28, 2026

T45. **Vanderbilt Center for Computational Systems Biology (CCSB) Retreat (Invited Talk),** "Bridging AI and Biology: Digital Twins, Foundation Models, and Agentic AI for Precision Brain Health", Vanderbilt University, Oct. 27, 2026

T46. **Vanderbilt Institute for Surgery and Engineering (VISE) Seminar Series,** "Multimodal AI for Precision Brain Health: From Digital Twins to Foundation Models and Agentic AI", Sep. 10, 2026

T47. **UF Institute of Aging,** Zoom, January 2022 "Artificial Intelligence in Cognitive Aging: Scalable and Personalized Intervention"

T48. **UF AI Town Hall,** Zoom, May 2021, "AI in Health at UF – Medical Image Analysis"

T49. **UF HiperGator Symposium**, Zoom, April 2021, "VCA-DNN: Neuroscience-Inspired Artificial Intelligence for Visual Emotion Recognition"

T50. **UF Informatics Institute Artificial Intelligence Seminar**, Zoom, March 2021, "AI in Precision Brain Dynamics"

T51. **UF College of Health and Human Performance Seminar**, Zoom, January 2021, "Artificial Intelligence in Predictive Health"

T52. **UF BMES Student Chapter Speaker Series**, Gainesville, FL. April 2019. "Big Biomedical Data for Tomorrow's Medicine"

T53. **UF ECE Machine Learning Workshop**, Gainesville, FL, January 2018, "Big Biomedical Data for Tomorrow's Medicine"

T54. **UF ECE Departmental Seminar Series**, Gainesville, FL. October 2017, "Big Biomedical Data for Tomorrow's Medicine"

T55. **McKnight Brain Institute,** Gainesville, FL. September 2017. "Big Biomedical Data for Tomorrow's Medicine"

T56. **FIU BME Wallace H. Coulter Foundation Lecture Series**, Miami, FL, September 2015. "Big Medical Data: Challenges, Opportunities, and Advances"

PATENTS

P1. **Ruogu Fang**, Skylar Stolte: System and Method of Digital Twins of Electrical Brain Stimulation, Filed 4/6/2025, 63/789,856.

P2. System and Methods for Predicting Perfusion Images from Non-Contrast Scans. Inventor(s): **Ruogu Fang**, Garrett Fullerton, Simon Kato. 63/395,561, published on Feb. 8, 2024. US-2024-0041417-A1.

P3. System and Method for Precision Dosing for Electrical Stimulation of the Brain. Adam Woods, Alexjandro Albizu, **Ruogu Fang**, Aprinda Indahlastari. PCT/US2021/043495, published on July 28, 2021. International patent published on Feb. 3, 2022. WO 2022/026573 A1. U.S. Patent No. 18/018,734. Patent application accepted on 3/22/2024.

P4. Systems And Methods for Image Denoising Via Adversarial Learning. Inventors: **Ruogu Fang** and Peng Liu. WO 2021/202170 A1. International patent. Published on Oct. 7, 2021. PCT/US2021/043635, Published on July 29, 2021.

P5. Multimodal CT Image Super-Resolution Via Transfer Generative Adversarial Network. Inventor(s): **Ruogu Fang**, Yao Xiao. U.S-2021-0272237-A1. Published on Sep. 2, 2021.

P6. Neural Network Evolution Using Expedited Genetic Algorithm for Medical Image Denoising. Inventors: **Ruogu Fang**, Peng Liu. Published on March 12, 2020. U.S. Utility Patent US 11,069,033 B2. Published on July 20, 2021.

P7. System and Method for Interactive Segmentation on Mobile Devices in a Cloud Computing Environment. **Ruogu Fang**, Leo Grady, Gianluca Paladini, Siemens Corporation, U.S. Patent No: US20130272587 A1, WO2012027259 A2, WO2012027259 A3. Published on March 1, 2012.

P8. A Machine Learning System and Method for Predicting Alzheimer's Disease Based On Retinal Fundus Images. Inventor(s): **Ruogu Fang** and Jianqiao Tian. T18201US001 (222107-8215). International Patent published on Dec. 2, 2021. WO 2021/243246 A1.

P9. Machine Learning for Predicting Parkinson's Disease Based on Retinal Fundus Images. Inventor(s): **Ruogu Fang**, Max Diaz. PCT/US2021/061182, published on Nov. 30, 2021.

P10. Prediction of Functional Motor Tasks with Dynamic Supervoxel Parcellations. Inventor(s): **Ruogu Fang**, Kyle See, Steven Coombes. PCT/US2022/081801, published on July 22, 2023.

P11. CFEA: Collaborative Feature Ensembling Adaptation for Domain Adaptation in Unsupervised Optic Disc and Cup Segmentation. Inventor(s): **Ruogu Fang**, Peng Liu, PCT/US2021/023678, published on March 23, 2021.

TEACHING

University of Florida, BME Department, Gainesville, FL

Instructor

- BME4931 Medical Artificial Intelligence (5.0/5.0)* **UF AI Course Award** F/23
- BME 3053C: Computer Applications for BME (Evaluation: F'20: 4.83/5) F'17-22
  - Final Project Spotlight: 2020, 2021, 2022
- BME 6938 Multimodal Data Mining* (S'19: 4.71/5, S'21: 4.92/5, S'23: 4.86/5) S'19, 21, 23
- NeuroAI T32 Machine Learning Workshop Summer'21

Guest Lecturer

- BME1008: Introduction to BME F'18,20, S'19-20
- BME4531: Medical Imaging F'18
- GMS6803: Data Science for Clinical Research F'19-22
- GMS6850: Foundations of Biomedical Informatics F'20,22
- ESI 6616: Data Analytics for System Monitoring F'22

Florida International University

Instructor

- CAP 4770: Introduction to Data Mining (Average Evaluation: 4.74/5) F'15-16
- CAP 5610: Machine Learning (Average Evaluation: 4.46/5) S'14-16

*Note: 'F' and 'S' denote Fall and Spring respectively. * indicates new course (series) I created.*

MENTORING

Current Postdoc Associates (2)

PD1. Peng Liu: 10/2023 - present (supported by NSF NCS award)

PD2. Wasif Khan: 10/2024 – present (supported by NSF SCH award)

Postdoc Alumni (1)

PD3. Diandra Ojo (Female, African American): 12/2023 – 08/2024

Current Ph.D. Students (5)

P1. Seowung Leem: 08/2021-08/2026

- International Student Achievement Awards Certificate of Outstanding Merit, University of Florida

P2. Joseph Cox: 08/2022-08/2027 (expected)

- NIH NIAAA T32 Fellow

P3. Junfu Cheng: 05/2024-08/2029 (expected)

P4. Zeyun Zhao, 05/2025 – 05/2030 (expected)

Ph.D. Alumni (total: 7)

P5. Zhimin Zong: 08/2022 – 05/2026
Dissertation: Evolutionary Mechanisms and Diagnostic Potential of Retinal Encoding: A Computational Approach

P6. Skylar E. Stolte (Female): 08/2020-08/2025

- Dissertation: Trustworthy Artificial Intelligence and Digital Twins for Precision Brain Modeling
- NIH F31 Ruth L. Kirschstein Predoctoral Individual National Research Service Award
- Graduate School Preeminence Award
- UF College of Engineering Attributes of Gator Engineer Award - Creativity
- Leighton E. Cluff Aging Research Award
- Pioneering Research HiPerGator Award
- UF BME Graduate Student Excellence Award
- Center for Cognitive Aging and Memory Dissertation Award

P7. Chaoyue Sun: 08/2020-08/2025

- Dissertation: Advanced Graph Neural Networks For Infectious Disease Dynamics
- Best Paper Award, The 17th International Conference on Health Informatics 2024
- International Student Achievement Awards Certificate of Outstanding Merit, University of Florida

P8. Kyle B. See: 08/2019 - 08/2024

- Dissertation: Interpretable Artificial Intelligence for Precision Health
- NIH TL1 Fellow, Graduate School Preeminence Award

- First Placement: Postdoc, ECE, University of Florida

P9. Tianqi Liu: 08/2019 – 01/2024

- Dissertation: Advanced Technologies for Video Coding
- Meta Industrial Foundation Fund
- First Placement: Senior Researcher, Tencent Americas, Palo Alto, California

P10. Peng Liu: 05/2016-12/2021

- Dissertation: Biology and Neuroscience Inspired Deep Learning
- UF Informatics Institute Fellowship
- Montreal AI & Neuroscience Conference Best Paper Award
- First Placement: Postdoc, Dartmouth College
- Current Position: Postdoc, University of Florida

P11. Yao Xiao (Female): 01/2017 – 08/2020

- Dissertation: Deep Learning for Multimodal CT Image Quality Enhancement and Radiation Exposure Optimization
- BMES Career Development Awardee, 2020.
- Commencement Student Speaker, University of Florida, College of Engineering, 2020.
- First Placement: Postdoc, MD Anderson, TX
- Current Position: Assistant Professor in Biomedical Informatics, Mayo Clinic, Rochester, MN

***Master's Alumni* (22)**

M1. Haodi Jiang, CISE, 2014-2015 (Ph.D. student at New Jersey Institute of Technology. Now Assistant Professor, Sam Houston State University)

M2. Jingan Qu, CISE, 2015-2017 (Now Data Scientist @ Cleveland Clinic)

M3. Daniel Parra, CISE, 2015-2016.

M4. Micheal Adeyosoye (African American): CISE, 2016-2017, Bridge to Ph.D. Fellowship.

M5. Skylar E. Stolte. BME, 2019-2020 (continue as a Ph.D. student in my lab)

M6. Jianqiao Tian: 08/2018-12/2020 (now PhD at Mayo Clinic)

M7. Shreya Verma (Female), BME, 03/2020 – 05/2021. (now at UIUC)

M8. Bhavin Soni, BME, 08/2019 - 05/2021 (First placement: Neuroimaging Engineer @Washington University in St. Luis. Current Placement: Data Scientist @ MD Anderson Cancer Center)

M9. Jiaqing Zhang, ECE, 03/2022-08/2022 (continued as PhD at UF ECE)

M10. Ziqian Huang, ECE, 03/2022-08/2022 (continued as PhD at UF BME)

M11. Ayesha Naikodi, 05/2022-05/2023 (continued as PhD at UF ECE)

M12. Kai Shen, BME, 05/2021-05/2023 (continued as PhD at UF IFAS)

M13. Jimmy Ossa, BME, 05/2021-05/2023 (now Software Engineer @ Infotech)

M14. Everett J. Schwieg, 08/2022 – 05/2024

M15. Fan Yang, 08/2022-05/2024 (Ph.D. student in NJIT)

M16. Divyansh Singh, CISE (continued as PhD at UF CISE)

M17. Prerna Venugopal, CISE, (now Software Engineer @ Suquino)

M18. Qiangeng Yang, ECE, (continued as PhD at UF ECE)

M19. Travis Koenig, 08/2023 - 12/2024 (First placement: Software Systems Engineer @ Arthrex)

M20. Pratyush Shukla, 08/2023 - 12/2024

M21. Zeyun Zhao, 08/2024-05/2025 (First place: Convert to PhD student in my lab)

M22. Daniel Rodriguez: 08/2022 – 12/2024

Current Postbacc Associate (1)

Gavin Hart (NIH GATORAADE Fellow)

Current Undergrad (7)

Grace Cheng, Jason Chen, Amy Lazarte, Samuel Zheng, Ritika Samanta, Sanandan Ojha, Sanjana Dundhigalla

NSF Graduate Research Fellowship Program (NSF GRFP) (3)

Maximillian Diaz, Daniel El Basha, Garrett Fullerton

NSF Research Experience for Undergraduates (REU) Mentee (10)

Mohammad Daniel El Basha, Kyle B. See, Akshay Mathavan, Akash Mathavan, Max Diaz, Garrett Fullerton, Simon Kato, Brian John Braddock, Jason Chen, Amy Lazarte

Past Undergrad (59)

Ahmet Bilgili, Harrison Santiago, Michael McGaha, Jacob Bleser, Edward Zhang, Hely Lin, Yiru Mu, John Braddock, Riddhi Banerjee, Ethan Smith, Justin Broce, Maria Cardei, Adeeb Rashid, Neeva Sethi, Peiwen (Riley) Gao, Abdul-Vehab Dozic, Eugene Li, Benjamin Arnold, Matthew Nguyen, Divyansh Singh, Joshua Lamb, Prerana Venugopal, Kyle See, Alvin Naiju, Akshay Ashok, Veronica Ramos, Thrisha Acharya, Minhyeok Wi, Kyle Douglas, Kevin Liu, Thomas Howland, Jordi Bardia, Brian John Braddock, Garrett Fullerton, Simon Kato, Gianna Sweeting, Michael Quentin McGaha, Keyur Patel, Nathan Barkdull, Sruthika Baviriseaty, Harrison Chaize Santiago, Nicolaas P de Jong, Sumanth Aluri, Max Diaz, Jason Chen, Mohammad Daniel El Basha, Akshay Mathavan, Akash Mathavan, Skylar Stolte, Paul Naghshineh, Sherman Ng, Michelle Mu, Rachel Peebles, Muhamed Maher, Isha Gupta, Rena Feng, Cynthia Liu, Rohan Shah, Yuanyuan Zhu, James-Bond Nguyen, Surya Karthikeyan Vijaylakshmi

UF AI Scholar / USP Program (9)

Akshay Ashok, Grace Cheng, Amy Lazarte, Jason Chen, Akshay Mathavan, Akash Mathavan, Maximillian Diaz, Garrett Fullerton, Gianna Sweeting

Secondary School Teachers (2)

Christian McDonald, Edda Rivera

University Minority Mentor Program (UMMP) (5)

Ariana Chang, Julinna Villarta, Liying Wu, Jordon Neal (African American), Joseph Wilkinson

Grad Society of Women Engineers (Grad SWE) (2)

Adriana Del Pino Herrera (BME PhD, Hispanic), Celene Cheddie (BME MS)

Student Science Training Program (SSTP) (3)

Srividya Vaishnavi Surampudi (now at Stanford University), Imaan Randhawa (now at UF), Aarushi Walia (now at Berkeley)

Integrated Product & Process Design (IPPD) (7)

Arjun Singh, Tuan Truong, Rafid Chowdhury, Jason Liang, Peter Salmon, Albert Li, Estefania Rodriguez Da Silva

Thesis & Dissertation Committees

Ph.D. Committee Chair

1. Yao Xiao, BME, Summer 2020
2. Peng Liu, BME, Fall, 2021
3. Tianqi Liu, ECE, Spring 2024
4. Kyle B. See, BME, Summer 2024
5. Chaoyue Sun, ECE, Summer 2025
6. Skylar E. Stolte, BME, Summer 2025
7. Seowung Leem, BME, Summer 2026 (expected)
8. Joseph Cox, BME, Summer 2026 (expected)
9. Zhimin Zong, ECE, Summer 2026 (expected)
10. Junfu Cheng, ECE, Summer 2029 (expected)
11. Zeyun Zhao, BME, Summer 2030 (expected)

MS Committee Chair

12. Travis Koenig, Spring 2025
13. Pratyush Shukla, Spring 2025
14. Shen Kai, BME, Spring 2022
15. Jimmy Ossa, BME, Spring 2022
16. Shreya Verma, BME, Spring 2021
17. Bhavin Soni, BME, Spring 2021
18. Jianqiao Tian, BME, Fall 2020
19. Skylar E. Stolte, BME, Spring 2020

Honor Thesis Committee Chair

20. Garrett Fullerton, BME, Spring 2023
21. Skylar E. Stolte, BME, Spring 2019
22. Kyle B. See, BME, Spring 2019
23. Maximillian Diaz, BME, Spring 2020

Senior Design Chair

24. Retinal Disease Diagnosis using Mobile Devices, Spring 2018: Daniel El Basha, Akash Mathavan, Akshay Mathavan, Kyle See, Skylar Stolte
25. Visual Amygdala Network with Predictive Coding for Emotion Recognition, Spring 2022: Michael Quentin McGaha, Eugene Li, Jordi Bardia, Keyur Patel.

Ph.D. Dissertation Committee Member

26. Xiaoshuang Shi (Major advisor: Mingzhou Ding), Biomedical Engineering, University of Florida, Spring 2019. Hashing for Large-Scale Pathology Image Retrieval and Classification
27. Ke Bo (Major advisor: Mingzhou Ding), Biomedical Engineering, University of Florida, Fall 2020. Neural Mechanism of Affective Scene Perception.
28. Guanhong Miao (Major Advisor: Samuel Wu), Biostatistics, August 2020. Techniques for Privacy Preserving Data Collection and Analysis
29. Camilo Valdes (Major advisor: Giri Narasimhan): Large Scale Human Microbiome Analytics, Florida International University, Spring 2020
30. Sudarat Tangnimitchok (Major advisor:Armando Barreto): Non-Intrusive Affective Assessment in the Circumplex Model from Pupil Diameter and Facial Expression Monitoring, Florida International University, Spring 2020
31. Terracino Brandon (Major advisor: Walter O'Dell), Medical Sciences, University of Florida, TBD
32. Trung Tran (Major advisor: Wesley Bolch), Medical Physics, University of Florida, Summer 2021. Pediatric Radiography Dosimetry for Radiation Epidemiology.
33. Cameron Berg Kofler (Major advisor: Wesley Bolch), Medical Physics, University of Florida, Summer 2021. The Development of a Newborn, Infant, and Toddler Mesh-Based Computational Phantom Library and a Pre-Computed Monte Carlo Dose Library for Computed Tomography.
34. William Moore (Major Advisor: S. "Bala" Balachandar), Aerospace Engineering, Spring 2021: Closing the Divide between Euler-Lagrange and Particle-Resolved Force Predictions in Dispersed Multiphase Flow
35. Bhargav Sriram Siddani (Major Advisor: S. "Bala" Balachandar), Mechanical Engineering, December 2022: Data-Driven Modeling for Enhanced Euler-Lagrange Simulations of Dispersed Multiphase Flows.
36. Nicole D. Evangelista, (Major Advisor: Adam Woods), Department of Clinical and Health Psychology, College of Public Health and Health Professions, Summer 2023： Investigating The Impact of Cognitive Training And Transcranial Direct Current Stimulation on Dorsolateral Prefrontal Compensatory Scaffolding.
37. Emanuel M. Boutzoukas, (Major Advisor: Adam Woods), Department of Clinical and Health Psychology, College of Public Health and Health Professions, TBD.
38. Alejandro Albizu, (Major Advisor: Adam Woods), Department of Neuroscience, College of Medicine, Fall 2023.
39. Sabyasachi Bandyopadhyay (Major Advisor: Parisa Rashidi), Biomedical Engineering, College of Engineering, Summer 2023.
40. Jorge Xavier Santiago-Hernandez (Major Advisor: Forest Masters), Civil Engineering, College of Engineering, Summer 2023.
41. Lihan Cui (Major Advisor: Mingzhou Ding), Biomedical Engineering, College of Engineering, Spring 2025
42. Yun Liang (Major Advisor: Mingzhou Ding), Biomedical Engineering, College of Engineering, Summer 2023.
43. Qiang Yang (Major Advisor: Mingzhou Ding), Biomedical Engineering, College of Engineering, Fall 2024.
44. Subhash Nerella (Major Advisor: Parisa Rashidi), Biomedical Engineering, College of Engineering, Fall 2023.
45. Jacob S. Salminen (Major Advisor: Dan Ferris), Biomedical Engineering, College of Engineering, TBD
46. Scott Siegel (Major Advisor: Parisa Rashidi), Biomedical Engineering, College of Engineering, Summer 2024.
47. Yongxu Zhang (Major Advisor: Shreya Saxena), Biomedical Engineering, College of Engineering, TBD
48. Changhao Xiong (Major Advisor: Mingzhou Ding), Biomedical Engineering, College of Engineering, TBD
49. Rob Chen (Major Advisor: David Vaillancourt), Biomedical Engineering, College of Engineering, TBD.
50. Jiaqing Zhang (Major Advisor: Parisa Rashidi), Electrical and Computer Engineering, College of Engineering, TBD
51. Tina Salehitorabi (Major Advisor: Kiley Graim), Computer Information Science and Engineering (CISE), College of Engineering, TBD
52. Zhimin Zong (Major Advisor: Jie Xu), Electrical and Computer Engineering, College of Engineering, TBD
53. Jared M. Baggett (Major advisor: Wesley Bolch), Medical Physics, University of Florida, TBD.
54. Hongxu Jiang (Major Advisor: Wei Shao), Electrical and Computer Engineering, TBD.

55. Fatemeh Baniasad (Major Advisor: Rachael Seidler), Health and Human Performance, Biobehavioural Science, (TBD)
56. Cameron G. Celeste (Major Advisor: Ivana Parker), Biomedical Engineering, (TBD)
57. Miguel Contreras (Major Advisor: Parisa Rashidi), Biomedical Engineering, (TBD)
58. Robert Joseph Dawson (major Advisor: Wesley Boch), Medical Sciences, (TBD)
59. Keaton August Micheal Reiners (Major Advisor: Perry Johnson), Medical Sciences, Medical Physics, (TBD)
60. Jessica Aneliza Shubin, (Major Advisor: Jada Lewis), Medical Sciences – Neuroscience, (TBD)
61. Myles Joshua Toledo Tan, (Major Advisor: Pinaki Sarder), Electrical and Computer Engineering (TBD)

Master Committee Member

62. Gang Qu, Computer Science, University of Florida, Spring 2018

Media Coverage

- A deep dive into ChatGPT moments in medicine - UF researchers dive into foundation models, UF Health News. https://fixel.ufhealth.org/2025/04/24/a-deep-dive-into-foundation-models-the-future-of-ai-in-medicine/, April 2025.
- Brain stimulation research is included in 2024 Coalition for Academic Scientific Computation (CASC) brochure.
- Brain/bio-inspired AI, Medical Imaging section of Computer Vision News on medical AI research. https://www.rsipvision.com/ComputerVisionNews-2023January/32/
- Artificial Intelligence for Precision Dosing to Prevent Dementia: *ABC Local10 TV, ABC WCJB TV, Alligator, UF Health News, Ivanhoe,* July 2021.
- Our eyes may provide early warning signs of Alzheimer's and Parkinson's. *The Washington Post*, by Elizabeth Anne Brown, February 27, 2021. https://www.washingtonpost.com/health/retiina-changes-in-early-alzheimers/2021/02/26/24c57bfa-6bba-11eb-9ead-673168d5b874_story.html
- Artificial intelligence for Alzheimer's and Parkinson's Disease early diagnosis from retinal imaging. *Forbes, RSNA, Yahoo, Medscape Medical News, American Journal of Medical Care, Big News Network, Diagnostics World News, SciTech Daily, Applied Radiology, The Hindu (India), EyeSmart (Australia), and Eminetra (South Africa), UF Engineering News, and 50+ International media.*
- ACM Future of Computing Academy, 2017.
- Professor Uses Computer Science to Reduce Patients' Exposure to Radiation From CT Scans, *ACM TechNews*, 2016.
- Researcher says head CT radiation dose can be reduced by 90 percent, *Healthcare Business*, 2016.
- FIU News: Professor uses computer science to reduce patients' exposure to radiation from CT scans, 2016.
- Facial recognition software spots family resemblance, *NewScientist*, 2012.
- Cambridge News: University of Cambridge Official News: Students from 'Cambridge of the East' take part in an exchange, 2009.

PROFESSIONAL ACTIVITIES

Journal Editorship

- Associate Editor, Medical Image Analysis (Impact Factor: 10.9), 2021-present
- Guest editor, VSI:Foundation Models for Computational Pathology in Medical Image Analysis
- Guest Editor, Current Opinion in Biomedical Engineering, Artificial Intelligence in Biomedical Engineering, 2025

- Topic Editor, Women in Brain Imaging and Stimulation, Frontiers in Human Neuroscience Journal, 2022
- Guest Editor, Computerized Medical Imaging and Graphics, Elsevier, 2015

Conference Organizing Committee and Track Chairs

- Track Chair, AI in BME Track, Annual Meeting of Biomedical Engineering Society (BMES), Orlando, FL. 2026.
- Organization Committee, Challenge Chair, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2027, Auckland, New Zealand.
- President, Women in MICCAI, the Medical Image Computing and Computer Assisted Intervention (MICCAI) Society, 2023-2025.
- Associate Editor, The Engineering in Medicine and Biology Conference, 2025
- Diversity Committee, Biomedical Engineering Society, 2024 -
- Co-host, Multimodal AI in Biomedical Data Analysis and Disease Diagnosis Special Session, the Annual Meeting of Biomedical Engineering Society (BMES), 2024, Baltimore, Maryland.
- Travel Award Committee, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024-2025.
- Co-host, Artificial Intelligence (AI) in Biomedical Engineering (BME) Education Special Session, the Annual Meeting of Biomedical Engineering Society (BMES), October 12, 2023.
- Award Committee, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2023.
- Program Committee, Area Chair, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2021-2024.
- Oral Session Chair, Integration Beyond Imaging, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2022
- Track Chair, Biomedical Imaging and Instrumentation Track, Annual Meeting of Biomedical Engineering Society (BMES), Orlando, FL. 2021.
- Session Chair, Optical Imaging I Session, Annual Meeting of Biomedical Engineering Society (BMES), Orlando, FL. 2021.
- Oral Session Chair, Image Reconstruction, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), September, 2021.
- Session Chair, Biomedical Imaging and Instrumentation Track, Deep Learning in Microstructural Imaging Session Chair, Annual Meeting of Biomedical Engineering Society (BMES), Philadelphia, PA, USA. October 16-19, 2019.
- Session Chair, Imaging Data Science, Processing Session, Modeling and Informatics Track, Image Processing and Analysis, Modeling, Data Science and Informatics Session, Annual Meeting of Biomedical Engineering Society (BMES), GA, Atlanta, USA. October 2018.
- Program Committee, 2019 KDD Workshop on Applied Data Science for Healthcare: Bridging the Gap between Data and Knowledge (DSHealth 2019), held along with KDD 2019, Anchorage, Alaska, USA. August 05, 2019.
- Session Chair, Annual Meeting of Society of Brain Mapping and Therapeutics, Los Angeles, CA. USA. April 2017.
- Program Committee Publicity Chair, IEEE International Conference on Machine Learning Applications, Miami, FL. USA. 2015.

- Co-Chair, 2nd International Workshop on Sparsity Techniques in Medical Imaging, in conjunction with International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Boston, MA, USA. September 2014

Proposal Panelist / Ad-Hoc Reviewer

- NIH Study Section, 2018-Present, 4 study sections, reviewed 20+ proposals.
- NSF Panel, 2015-Present, 11 panels, reviewed 70+ proposals.
- Parkinson's UK, 2025-Present.
- Swiss National Science Foundation, 2023-Present.
- Wellcome Funding, United Kingdom, 2021-Present

Reviewer

1. The Lancet (IF=59.102)
2. Nature Biomedical Engineering (IF=26.6)
3. Nature Neuroscience (IF=20)
4. Nature Machine Intelligence (IF=18.8)
5. Science Advances (IF=11.7)
6. IEEE Transaction on Pattern Analysis and Machine Intelligence (IF=17.730)
7. IEEE Reviews in Biomedical Engineering (IF=17.20)
8. IEEE Transaction on Neural Networks and Learning Systems (IF=11.683)
9. Journal of Information Fusion (IF=10.716)
10. Medical Image Analysis (IF=8.545)
11. Journal of Movement Disorders (IF=8.324)
12. IEEE Transaction on Medical Imaging (IF=7.861)
13. IEEE Transaction on Biomedical Engineering (IF=4.491)
14. ACM Computing Survey (IF=6.131)
15. Journal of Pattern Recognition (IF=5.898)
16. IEEE Transaction on Image Processing (IF=5.071)
17. IEEE Journal of Biomedical and Health Informatics (IF=4.217)
18. Nature Scientific Reports (IF=4.011)
19. Frontiers in Neurology (IF=3.552)
20. NeuroComputing (IF=3.317)
21. Journal of Sensors (IF=3.031)
22. Physics in Medicine and Biology (IF=3.030)
23. Plos One (IF=2.776)
24. Journal of Structural Engineering (IF=2.528)
25. IEEE Transaction on Multimedia (IF=5.452)

26. Neuroradiology (IF=2.504)
27. Journal of Visual Experiments (IF=1.530)
28. Machine Vision and Applications (IF=1.103)
29. International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)
30. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
31. IEEE International Conference on Computer Vision (ICCV)
32. IEEE International Conference on Image Processing (ICIP)
33. IEEE International Symposium on Biomedical Imaging (ISBI)

Society Mentorship

- Mentor, National Academy of Inventors Global Academy Inventors Networks (GAIN).
- Mentor, AnitaB.org Global Women Technologist and Mentorship Program (mentored 8 women in the technology field globally for 3 months)
- Mentor, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2023.

University and Departmental Service

- Chair, BME Inclusive Excellence Committee, 2023-2024.
- Co-host, AI Affinity Group, Hebert Wertheim College of Engineering, University of Florida, 2023-
- Chair, BME Engagement, Outreach and Public Relations Committee, 2021-2022
- MSAIS & MSADS Curriculum Committee, Hebert Wertheim College of Engineering, 2023-
- UF Health Science Center AI Search Committee, 2020-2021
- BME Engagement, Outreach and Public Relations Committee, 2023-2024
- BME Graduate Committee, 2023-2024
- BME Space Committee, 2023-2024
- BME Tenure and Promotion Committee, 2022-2024
- BME Space Committee, 2023-2024
- BME Executive Committee, 2021-2023
- BME Tenure and Promotion Committee, 2022-2024
- BME Faculty Search Committee, 2019-201
- BME Undergraduate Program Committee (UPC), 2017-2021
- BME Seminar Committee, 2017 – 2020

REFERENCES

James Duncan
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